

Pathway for a fully renewable power sector of Africa by 2050: Emphasising on flexible generation from biomass

Theophilus Nii Odai Mensah^{*}, Ayobami Solomon Oyewo, Dmitrii Bogdanov, Arman Aghahosseini, Christian Breyer

LUT University, Yliopistokatu 34, 53850, Lappeenranta, Finland

ARTICLE INFO

Keywords:

Africa
Dispatchable renewable energy
Energy transition
Modern biomass application
Sustainable biomass potential
100% renewable energy

ABSTRACT

The climate crisis coupled with energy poverty and injustice in Africa requires strategic measures to proffer sustainable solutions. A transition to a defossilised power sector is obtained for Africa by 2050, evolutionarily developed to fulfil the Paris Agreement. With the LUT Energy System Transition Model, the role of biomass in a least-cost power sector for Africa is investigated from 2020 to 2050 in five-year time steps. Techno-economic assumptions of employed technologies are applied. The study considers the current generation portfolio, life-span of power plant technologies, and the current and future electricity demand to ascertain the best generation mix by 2050. Biomass potential of 334 TWh (17.5%) of the total African biomass potential of 1911 TWh is applied to the African power sector to develop two Best Policy Scenarios: a scenario without bioenergy (BPS-1) and a scenario with bioenergy (BPS-2). The results indicate a positive impact of bioenergy on the power sector: The total generation, total installed capacity, grid utilisation, levelised cost of electricity, and storage output decreased by 8.6%, 8.4%, 12.9%, 10.2%, 47.1% by 2050 respectively in BPS-2 relative to the BPS-1. In addition, flexible generation from biomass offers balancing to enhance variable renewable energy resources integration onto the power sector.

1. Introduction

With respect to the problems of energy injustice, climate change, global warming, and pollution, the global energy sector is in need of a complete restructuring [1–4]. The global energy sector is responsible for about 73.2% of all anthropogenic greenhouse gas (GHG) emissions due to the massive dependence on fossil fuels for energy supply [5]. This requires a paradigm shift to renewable energy (RE) resources that are environmentally benign and sustainable [6–9]. Economic development can be achieved sustainably by investing in sustainable renewable energy technologies (RET), thereby breaking the economic growth and GHG emissions nexus and ensuring climate action. Investing in RET will limit fossil fuel use in the power sector and address climate change vulnerabilities [1,10]. Studies have shown that although Africa contributes just a minuscule (4%) of the annual global GHG emissions, the continent is highly susceptible to the negative impacts of global climate change [11].

The African continent is the second largest in the world with a population of 1.3 billion in 2019 [12]. It consists of 54 countries and an

area of over 30 million square kilometres [13]. Paradoxically, although the continent is endowed with energy resources, access to electricity is poor due to inadequate power generation and transmission infrastructures [14]. Currently, the electricity access rate of Africa is as low as 47.9% compared to the global average of 90% in 2019 [15]. The average annual electricity consumption per capita in sub-Saharan Africa (SSA) is as low as 150 kWh/capita [14–16], culminating in the region accounting for just 3% and 6% of the global electricity and energy demand respectively in 2018 [14]. Consequently, the rate of economic growth (3.2%, 3.4%, 3.7%, in 2018, 2019, and 2020 respectively) is hampered [17]. The average gross domestic product (GDP) per capita in SSA was as low as 1629 USD compared to the global average of 11,395 USD in 2019 [18].

Furthermore, demand for electricity is growing steadily, and electricity deficit is anticipated to increase owing to urbanization, population growth, and industrialisation [19,20]. In response to these challenges, new generation capacities, and transmission infrastructure are being added to achieve universal access to electricity [21–23]. However, these capacity additions will undoubtedly increase the GHG emissions if the current trend of power generation persists. Currently,

^{*} Corresponding author.

E-mail address: theophilus.mensah@lut.fi (T.N.O. Mensah).

<https://doi.org/10.1016/j.renene.2024.121198>

Received 17 March 2022; Received in revised form 13 August 2024; Accepted 15 August 2024

Available online 15 August 2024

0960-1481/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Abbreviation			
AD	anaerobic digestion	LCOT	levelised cost of transmission
BPS	Best Policy Scenario	LHV	lower heating value
BAU	Business as usual	MSW	municipal solid waste
CAPEX	capital expenditure	OCGT	open cycle gas turbine
CCGT	combined cycle gas turbine	OPEX	operational expenditure
CO ₂	carbon dioxide	PV	photovoltaic
CHP	combined heat and power	PHES	pumped hydro energy storage
DAC	direct air capture	PtG	power-to-gas
FAO	Food and Agriculture Organization	TES	thermal energy storage
FLH	full load hours	DH TES	district heating thermal energy storage
FSS	faecal sewage sludge	A-CAES	adiabatic compressed air energy storage
GT	gas turbine	RE	renewable energy
GHG	greenhouse gas	RoR	run-of-river
HVDC	high voltage direct current	RPR	residue to product ratio
IEA	International Energy Agency	SNG	synthetic natural gas (e-methane)
IRENA	International Renewable Energy Agency	SDG	Sustainable Development Goals
LCOC	levelised cost of curtailment	SSA	sub-Sahara Africa
LCOE	levelised cost of electricity	TWh	Terawatt-hour
LCOS	levelised cost of storage	VRE	variable renewable energy
		WACC	weighted average cost of capital

Africa generates almost 80% of its electricity from fossil fuels [24]. A practice that has a negative economic and environmental impact on the continent due to volatile world oil prices and pollution respectively [4]. In this regard, the African continent must seek a power sector reform that will ensure the realisation of a secure, resilient and climate benign power sector that conforms with the global climate goals in meeting the current and future electricity demand [25]. Additionally, Oyewo et al. [26] concluded that many African countries are greenfields for developing RE systems due to the challenges of increasing energy demand, lack of supply, and climate change mitigation and adaptation. Furthermore, the Agenda 2063 of the African Union Commission's (AUC) stipulates the need to provide clean energy for all Africans by 2063 [27].

In this regard, the continent's huge RE resources potential, coupled with rapid cost reduction of RET, especially solar PV, is a key motivating factor for rapid investments in utility-scale renewables for an urgent energy sector revolution in Africa [28–31]. Effective planning and wise utilisation of the untapped, huge RE resources in Africa will help to alleviate current and future energy crises and, at the same time, meet climate action goals [26].

To establish the knowledge gap, a comprehensive literature review on 29 peer-reviewed journal articles and two reports was conducted on the contribution of RE in the transition of the power sector of Africa to 100% renewable electricity. Reviewed journal articles consist of 13 overnight and 15 transition studies. These studies are extracted from the known 100% RE studies for Africa [32]. Results of the reviewed studies indicate the technical feasibility and economic viability of a RE transition in Africa. Solar and wind energy resource have been projected to supply the bulk of the power demand by most of the studies. This study is particularly focused on a robust hourly resolution for the power transition of Africa and the role of bioenergy in complementing the bulk power supply from solar PV, wind power and hydropower generators. Table 1 below presents the list of published studies on fully RE systems in Africa.

The reviewed studies on 100% RE systems show consistent evidence that VRE resources, especially solar PV and wind power will supply the bulk of the future electricity in Africa. These studies focused predominantly on SSA, West Africa, North Africa, individual African countries, and a recent study for the entire African continent. Notably, most of the studies that focused on SSA were all overnight studies that employed flexibility and balancing options other than bioenergy. Additionally, these studies lack robust hourly resolution power transition modelling.

However, Oyewo et al. [41] focused on the entire African continent in an hourly resolution transition for the complete energy system but also employed e-fuels, sector coupling, and other storage options for system balancing. None of the reviewed studies focused on an hourly resolution transition for the power sector only for Africa. Thus, there is a lack of robust scientific study on hourly resolved power sector transition in sufficient high regional resolution, and the integration of VRE resources into the power system for a stable and reliable power supply in Africa considering their variable characteristics. There is a gap of understanding the hourly resolution power transition and the role of dispatchable bioenergy and its impact on storage options in Africa as can be observed in the reviewed studies.

Therefore, to fill this research gap, a novel and comprehensive transition analysis of the power sector is carried out from 2020 to 2050, in a full hourly and high geo-spatial resolution for entire Africa. Additionally, as shown by Mensah et al. [53] on the role of bioenergy in providing flexible generation for balancing the power grid for the case of Ghana, this study also estimates the potential of waste and residue biomass (sustainable biomass) of individual African countries and investigates its potential for offering flexible generation to provide balancing to the variation in generation of inflexible VRE for the system to integrate large shares of VRE resources in Africa [53,63,64]. This study is the first of its kind for Africa.

Although the entire African energy system relies heavily on bio-energy (45.3% of total demand in 2018), current biomass electricity generation is low at 2 TWh, which is less than 1% of the entire electricity generation of 837 TWh by 2018 [65,66]. Fig. 1 show the contribution of biomass to the African energy and power sector.

The sections of the paper are arranged as follows; Section 2 discusses the biomass potential in Africa, section 3 describes the methods, sections 4 present the results, section 5 discusses the results, and conclusions are present in section 6.

2. Biomass potential in Africa

Biomass resources in Africa can be classified into the following categories: a) forestry biomass, b) agriculture, and c) waste and residues [53]. Examples of biomass include woody biomass, fruit stones, herbaceous, livestock manure, faecal sewage sludge (FSS), crop residues, forest residues, and municipal solid waste (MSW) [53].

Forestry biomass includes forestry woody products; agriculture

Table 1

A brief review of fully renewable energy systems studies in some regions and countries of Africa.

Author	Sector	Transition	Overnight	Remarks
Global				
Greenpeace [33]	All	X		This study illustrates that the RE share in the African power sector could increase to 31%, 65%, and 100% by 2020, 2030, and 2050 respectively.
International Energy Agency (IEA) [34]	All		X	In their New Policies Scenario, This IEA study reveals that RE can contribute almost 55% of Africa's electricity by 2040.
Teske et al. [35]	All		X	This study investigates the possibility of achieving the Paris Agreement by analysing two global non-overshoot pathways (+2 °C and +1.5 °C) by using normative scenarios to show the impact of energy efficiency measures, and RE utilisation and the results for Africa show that solar, biomass, wind, and e-fuels provide almost 44%, 20%, 19%, and 17% respectively of the energy supply in the +1.5 °C scenario by 2050.
Breyer et al. [36]			X	This study performs a global 100% RE system analysis based on an overnight approach which captured SSA, and the results indicate that by 2030, solar and wind energy will emerge as the bulk electricity provider for SSA.
Jacobson et al. [9]	All		X	This study analyses a global (143 countries) 100% RE system based on an overnight approach. The results show that transitioning to a fully renewable power sector is not only technically feasible and economically viable but will also create more jobs, decrease air pollution, reduce energy poverty in Africa, and subsequently reduce global warming.
Bogdanov et al. [37,38]	All		X	These studies conclude that SSA can satisfy its electricity and energy demand with 100% RE by 2050. Solar PV alone will provide 82% of the total supply for the power sector and all energy demand.
Ram et al. [39,40]	All		X	These studies conclude that 5.5 million and 12.2 million jobs can be created in SSA for the power sector and all energy supply by 2050, respectively.
Africa				
Oyewo et al. [41]	All	X		This study simulates the complete energy transition from 2020 to 2050 and concludes by suggesting that energy policy in Africa should have solar PV at its core and flexibility options such as grid interconnections, storage, and e-fuels are needed to integrate large shares of VRE in the African energy system.
SSA				
Barasa et al. [42]	Power		X	This study investigates a cost-effective power sector for SSA by 2030 by analysing an area-wide and integrated area-wide scenario based on an overnight approach and concludes that by 2030, a 100% RE system is both technically feasible and economically viable and that solar and wind energy will supply the bulk of the electricity.
Oyewo et al. [43]	Power		X	This study reveals the potential of RE to supply 100% renewable electricity for SSA, while discussing the limited economic possibility of Grand Inga.
West Africa				
Oyewo et al. [44]	Power	X		This study shows that it is not only possible to achieve a fully renewable power sector for Africa by 2050, but it will also be the least cost pathway.
North Africa				
Aghahosseini et al. [45]	Power		X	This study indicates that renewables can supply sufficient energy to cover the electricity demand in the Middle East and North Africa (MENA) region by 2030 on a least cost level of 40.3–52.8 €/MWh as compared to a BAU scenario cost of 118.6–149 €/MWh.
Country				
Timmons et al. [46] Mauritius	Power		X	The study demonstrates Mauritius' ability to supply its electricity demand in 2019 with renewable electricity at a same cost level to conventional generation.
Timmons et al. [47] Mauritius	Power		X	This study discusses the economic impact of energy storage units in the Mauritius 100% RE power sector by 2040 and concludes that meaningful cost reduction can be achieved via PHES systems.
Bouckaert et al. [48] Réunion Island	Power	X		This research discusses the role of energy storage in the Réunion island power sector by 2030 for sustainable power supply and the result shows that with sufficient investment in energy storage facilities, the island could meet its electricity demand with 100% RE with a high share of variable renewable energy (VRE) plants of about 50% without interruption in transmission.
Drouineau et al. [49] Réunion Island	Power	X		This research concludes that by 2030, Réunion island can achieve electricity self-sufficiency with available RE resource potential and provides the pathways for achieving such power generation mix.
Maïzi et al. [50] Réunion Island	Power	X		A fully RE power sector is achieved for Réunion island until 2030. It is shown that kinetic energy influence is vital in the power systems ability to handle sudden disturbances.
Selosse et al. [51] Réunion Island	Power	X		This article concludes that by 2030, Réunion Island can achieve a 100% RE power sector and that 50% of the island's power demand can be achieved via biomass electricity to replace coal generation.
Selosse et al. [52] Réunion Island	Power	X		This study discusses some important policies that can aid to revolutionise the power sector of Réunion Island to attain 100% RE by 2030 and conclude that biomass appears to be the most promising substitute for fossil fuels.
Mensah et al. [53] Ghana	Power	X		A 100% RE power sector can be achieved for Ghana. The research shows that bioenergy can play an essential role in the power sector by offering grid balancing responsibilities to offset the variability of VRE generation.
Oyewo et al. [54] Nigeria	Power	X		A 100% RE power can be obtained for Nigeria. The study concludes that PV-battery systems can supply the bulk of Nigeria's power sector by 2050.
Tambari et al. [55] Nigeria	Power	X		The study suggests that by 2050, RE resources can provide the power demand of Nigeria and that 100% renewable electricity is more cost-effective than the current generation portfolio by 2050. In addition, 1.54 million jobs can be created with a RE power sector in Nigeria.
Oyewo et al. [56] South Africa	Power	X		This study reveals that by 2050, a cost-effective, environmentally friendly, and a fully RE power sector can be achieved in South Africa.
Ferreira et al. [57] Cape Verde	Power	X		This research concludes that a 100% RE power sector for Cape Verde may result in an increased total system cost. However, it would reduce CO ₂ emissions.
Nordman et al. [58] Cape Verde	Power			This review explores the various options available for Cape Verde to achieve a 100% RE electricity target and concludes that it is possible for Cape Verde to achieve this target. However, the transition will be capital intensive and will require an investment of 1 bUSD.

(continued on next page)

Table 1 (continued)

Author	Sector	Transition	Overnight	Remarks
Manjong et al. [59] Cameroon	All	X		This study reveals that solar PV, hydropower, and sustainable bioenergy can supply the energy needs of Cameroon by 2050 and that there are no major barriers to a fully RE system in the country.
Oyewo et al. [60] Ethiopia	All	X		This research reveals that a RE dominated energy system of Ethiopia is least costing, least greenhouse gas emitting, and can create more jobs than the existing energy system.
Breyer et al. [61] Morocco and Algeria	Power	X		A fully RE power transition in this study reveals that electricity demand in Algeria and Morocco can be supplied with available RE resources in the region by 2050. In Algeria, solar PV, and wind power supply 77% and 21% respectively of electricity demand by 2050. Similarly, in Morocco, solar PV and wind supply bulk of the electricity demand by the target year 2050.
Schinko et al. [62] Morocco	Power		X	This study analyses different scenarios for the Moroccan power sector and concludes that de-risking investments in RE technologies could lead to a cost-competitive fully renewable based electricity generation for Morocco.

All sectors include power, transport, and industry.

Overnight scenario considers only a particular year.

Transition scenario considers a modelling from an initial year to a target year in 5 years step.

biomass includes mainly energy crops while waste and residue biomass include livestock manure, faecal sewage sludge (FSS), crop residues, forest residues, and municipal solid waste (MSW).

Africa contributes some 17% of the global forest, translating to 650 million hectares of forest cover. Africa has the most biodiverse form of forest covering about 22% of the total land area of Africa. The Congo basin, Western and Central Africa account for the largest share of the continent's forest [58]. Forestry biomass for energy such as fuel wood, pine bark, poplar, and wood pellets are from trees harvested for energy purposes except forest residues which can be harvested from both industrial round wood and wood fuel. Primary forest residues such as limbs, stumps, stems, tree branches and secondary forestry residues such as bark, and sawdust are used traditionally in Africa [68].

Also, agricultural crop residues which include stalks, husks, cobs, shell, and straw [69] are used in rural Africa for cooking, and heating purposes and the leftovers are burned in open fields [70]. Furthermore, management of FSS, manure, and MSW in SSA is poor as sufficient treatment facilities are not available. Nonetheless, the generation of this type of waste is increasing [71,72]. Waste treatment technologies such as anaerobic digestion (AD) plants, waste-to-energy power plants [73–76], recycling plants, and composting plants are very limited in capacity or not available [77,78] which results in open dumping and uncontrolled landfilling of waste as disposal methods. Somorin et al. [79] have stated the potential of energy recovery from MSW via incineration in Africa. Also, Scarlat et al. [80] reported that MSW generation in Africa has an energy potential of 312.5 TWh and 610.6 TWh in 2012 and 2025 respectively, if all waste generated in urban Africa are collected.

Several studies have shown the biomass potential of Africa ranging from 0 TWh/a – 6820 TWh/a for all biomass categories, which includes energy crops (0 TWh/a – 3861 TWh/a), forestry (0 TWh/a – 1500 TWh/a), and residue and waste (3 TWh/a – 1459 TWh/a) [81–83].

It has been established that over 700 million of the African population uses traditional biomass for heating, household cooking, and other economic activities [84]. Currently, the use of animal dung, charcoal, tree leaves, fuel wood, and agricultural residues for heating purposes,

which is termed as traditional biomass is prevalent in SSA. Traditional biomass partially contributes to forest depletion, soil degradation, loss of productive time, especially of women who normally collect the fuel-wood from the forest, and eventually, poverty due to loss of productive time [84–86]. This research work is conducted to address the possible alternative of traditional biomass by using sustainable biomass for electricity supply in Africa to enhance the integration of VRE resources in the power sector. Currently, biomass for cooking and heating in Africa demand high shares of biomass due to a low conversion efficiency (10%–20%) [87]. According to Keiner et al. [88], SSA contributes 48.3% (2830 TWh) and 60% (0.372 TWh) of the total global biomass for cooking, estimated to be 5860 TWh and 0.62 TWh in 2012 and 2045, respectively. Musonda et al. [89] have shown that an optimum GHG abatement benefit is achieved when biomass is allocated to all the sectors of the bioeconomy including biochemicals, than when allocated to only power, building, transport, or industrial sectors. Therefore, in this research, a large share of the biomass potential is reserved to satisfy demand in the heating sector including cooking, biofuel for transport, and biomass to chemicals.

3. Method and approach

The methods applied in the study are discussed in this section. The bioenergy potential method [53] and the LUT Energy System Transition Model (LUT-ESTM) [37] are applied. The bioenergy estimation result is applied as input data to LUT-ESTM to simulate the bioenergy impact on the African power sector.

3.1. Sustainable biomass estimation approach

Biomass considered for this study includes only waste and residues for sustainability reasons. For estimating the sustainable biomass potential, energy crops are excluded for environmental sustainability reasons. A sustainability constraint of 70% is applied to the current production level of forestry biomass, while forest residues, crop residues, FSS, and manure utilisation for energy are set at 35%, 35%, 80%,

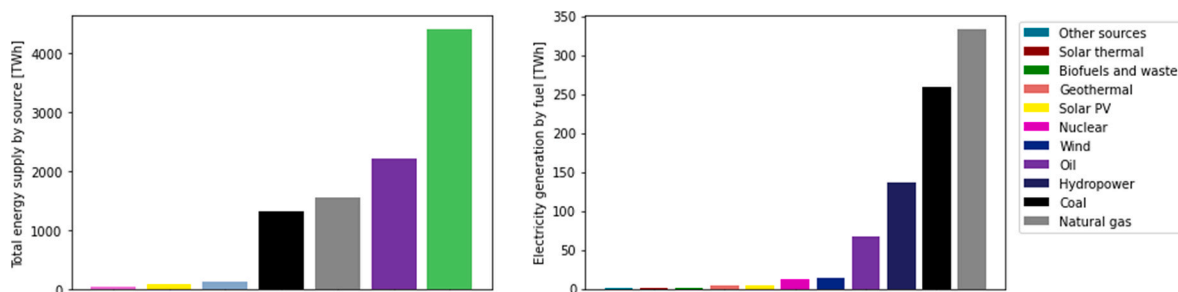


Fig. 1. Total primary energy supply by fuel (left) and electricity generation by fuel (right) in Africa in 2018 [66].

and 80%, respectively. MSW is treated with mass combustion for energy recovery. The biomass potential used in the model is grouped into three categories:

- solid waste, including used wood, and MSW (excluding food waste).
- solid residue, including coconuts, sugar cane, straw, and forestry residues.
- biogas including livestock manure, bio-food waste, and FSS.

Data for wood and crop production are obtained from FAO [90,91]. The residue determinant parameter, residue-to-product ratio (RPR) [92] and the use factor [92] is used to obtain forest and crop residue available for energy. Lower heating values (LHV) of forest and crop residues are obtained from Ref. [92]. The moisture content of wood is obtained from Ref. [92]. Livestock population is obtained from the FAO database [93]. Manure per head of livestock [94] and a use factor of 80% is assumed to calculate manure available for energy. The quantity of FSS available for energy is calculated based on the African population [95], faeces per capita [96], and a use factor of 80% is assumed. The methane content of biogas is obtained from Ref. [97]. MSW availability for energy is calculated based on the average MSW generation per capita in Africa [98], population [95], and recycling rate [98]. The cost for biomass (forest and crop residues) was based on [99]. For MSW cost calculation, a gate fee was derived based on the individual country's GDP. The cost for biogas feedstock is assumed to be zero. Detailed equations of the biomass method are described in Ref. [53]. Renewable resources' full load hours (FLH) and generation profiles can be found in the Supplementary Material (Tables A12–A13 and Fig. A3). Fig. 2 below describes the flow chart of the bioenergy estimation method.

3.2. LUT Energy System Transition Model representation

The power sector modelling is done with the LUT-ESTM described in Ref. [37]. The 54 African countries are merged into 24 regions forming



Fig. 3. The 24 regions of Africa. MA: Morocco; DZ: Algeria; TN: Tunisia; LY: Libya; EG: Egypt; MW: Mauritania, Western Sahara; WW: Senegal, Sierra Leone, Gambia, Guinea Bissau, Cape Verde Islands, Guinea, Liberia; MLI: Mali; WS: Benin, Cote D'Ivoire, Burkina Faso, Togo, Ghana; NER: Niger; NGA-S: Nigeria South; NGA-N: Nigeria North; SER: Sudan, Eritrea; ETH: Ethiopia; SOMDJ: Djibouti, Somalia; KENUG: Kenya, Uganda; TZRB: Tanzania, Rwanda, Burundi; TCD: Chad; CAR: Congo, Sao Tome and Principe, Guinea Equatorial, Gabon, Cameroon, Central African Republic; COG: Democratic Republic of Congo; SW: Angola, Namibia, Botswana; ZAFLS: South Africa, Lesotho; SE: Malawi, Mozambique, Zambia, Zimbabwe, Swaziland; and IOCE: Seychelles, Mauritius, Madagascar, Comoros Islands, and Mayotte.

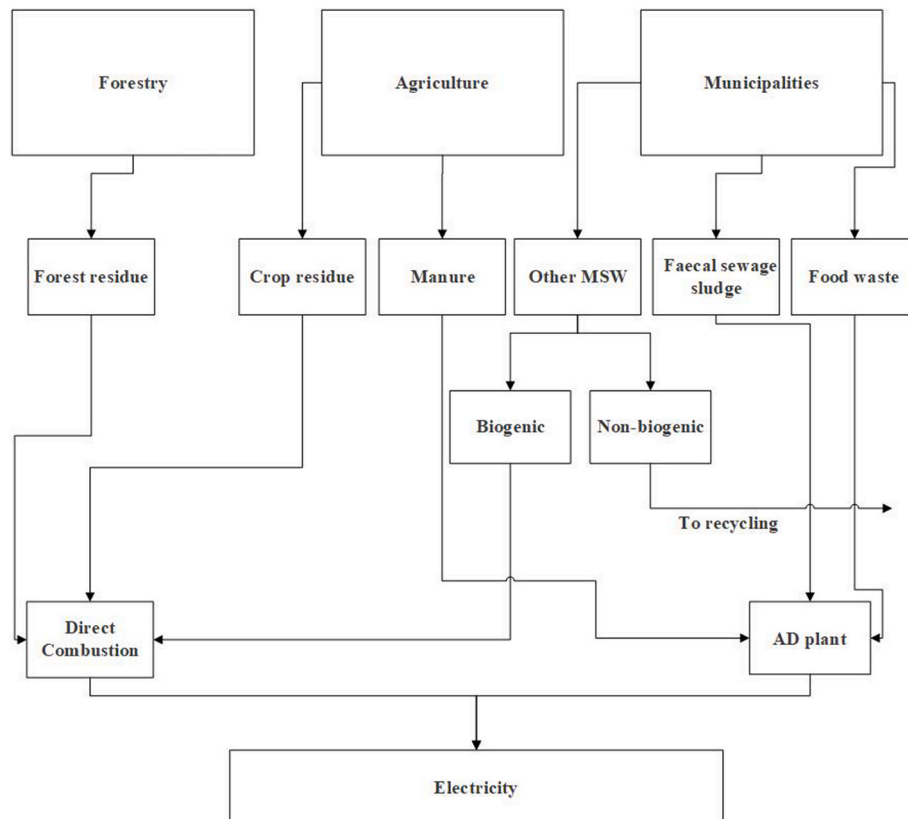


Fig. 2. Flow chart of the sustainable bioenergy estimation method [53].

24 nodes in the model and further grouped into 5 macro-regions as shown in Fig. 3.

The prime goal of the LUT Energy System Transition model is to achieve a good mix of RE resources by optimizing the sum of the yearly cost of different technologies, energy generation and generation ramping to minimize the yearly system cost thereby reducing the end-use electricity cost. The model employs a multi-nodal, linear optimization technique to ensure a balance between power demand and supply for every hour of a complete year under defined operational constraints and assumptions. Additionally, distributed self-generation and consumption of industrial, residential, and commercial electricity consumers (prosumers) is embedded in the model through the installed capacities of batteries and rooftop PV systems. When prosumers produce excess electricity, it is sold to the national grid for 0.02 €/kWh, and prosumers in turn buy from the national grid when self-production is low. The total annualised cost of the prosumers is the sum of self-generation cost and grid electricity consumption cost minus electricity sold to the grid. The model works under the following defined constraints. Firstly, after 2020, fossil fuels based, and nuclear power plants are not permitted to be installed due to sustainability reasons. However, gas turbines can be installed due to fuel switching possibilities. Secondly, hydropower plants are refurbished every 35 years. Thirdly, the annual growth rate of RE capacity is kept at or below 4% to prevent the model from deploying RE technologies too quickly. Fourth, the maximum prosumer demand is limited to 20% of the total demand. In the first step, prosumers are assumed to be 3% of the consumers. The prosumer shares progress in the subsequent periods in a step wise manner to 6%, 9%, 15%, 18%, and 20%. Lastly, bioenergy utilisation is regulated by allowing 33%, 66%, and 100% potential utilisation by 2020, 2025, and 2030 respectively. By this, the rolling out of bioenergy technologies is controlled.

One of the benefits of the used model design is applying a hierarchical approach to reduce the complexity of the simulation, which allows modelling the system with high temporal, technological and regional resolution or decreasing simulation time. As a first step, the model is applied to a reduced number of larger regions, named macro-regions, which are because of merging the regions described in Fig. 3, aggregating energy demand, existing capacities, and the RE potential. To reduce the copperplate effect, i. e., the assumption that power can flow from any generation point to any demand point without constraints, of the RE profiles, the macro-regions are calculated as a weighted average of the defined regional RE profiles considering costs, electricity generation and distances between regions. The results of the first level simulations provide information on power flows among the

macro-regions, which is later used as input information for the second level simulations. On the second level of the modelling, each macro-region is represented as set of interconnected regions with specific energy demand, capacities, RE potential and profiles. Then each macro-region is simulated individually. The power flow profiles from the previous step are considered and added to the power balancing equation of the regions through which the macro-regions were interconnected. At the last step, the results of all the macro-region simulations are merged, considering the power flows among macro-regions from the first step. As a result, the operation of the system is simulated as if all considered regions were interconnected with power grids. This approach allows for the reduction of number of variables in each simulation by 3–5 times, depending on the size of the macro regions, and consequently reduces the simulation time. Model information and equations can be found in Bogdanov et al. [37,100], and design information can be found in the Supplementary Material (Fig. A1). Fig. 4 present the flow chart of LUT-ESTM.

3.3. Applied technologies

The different technologies employed in the model for analysing the role of sustainable biomass in the African power sector are electricity generation, including solar PV and concentrating solar thermal power (CSP), wind turbines, hydropower, gas turbines (renewable gas as fuel, i. e. biomethane or e-methane), biomass power plants, and geothermal power plants; power transmission, including high voltage alternative current (HVAC), high voltage direct current (HVDC); storage units, including battery [101], adiabatic compressed air energy storage (A-CAES) [102], pumped hydro energy storage (PHES) [103], and power-to-gas (PtG, e-methane) storage [104] consisting of CO₂ direct air capture [105], electrolyzers, methane and gas turbines. Distribution and transmission grid losses, and electricity load profiles were taken from Sadoskaia et al. [106] and Toktarova et al. [107] respectively. More information about the model is found in the Supplementary Material (Fig. A2).

3.4. Potential of renewable energy

Reliable energy resource estimates are key prerequisites for the planning of a fully renewable power sector. This research considered biomass, wind, hydro, solar, and geothermal for power supply. To ensure long-term energy security for all African countries, it is essential to consider several RE resources capable of satisfying current and future

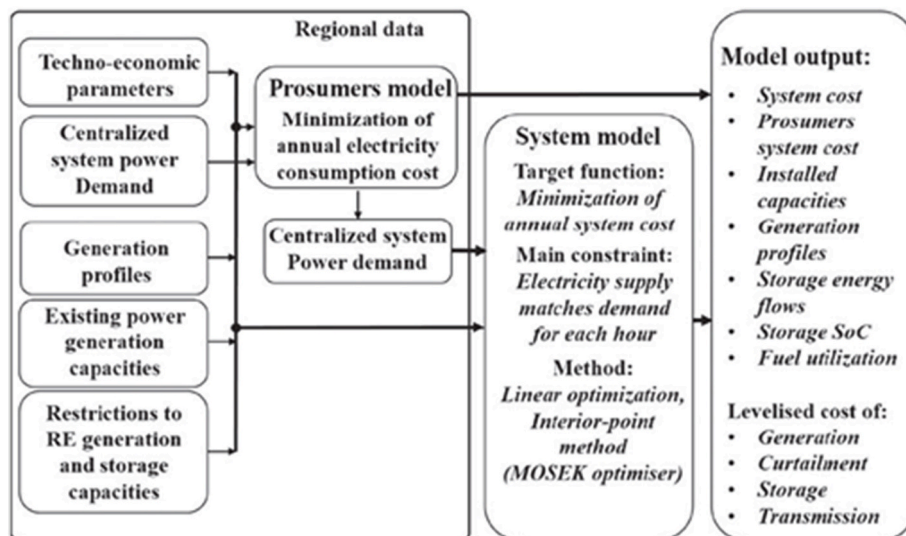


Fig. 4. Flow chart of the LUT Energy System Transition Model [37].

demand. Based on a German Aerospace Centre [108] reprocessed NASA data [109,110], the generating profiles for solar and wind resources are calculated according to Refs. [111,112]. The feed-in profile for hydropower is estimated according to the method described in Ref. [113]. The method of calculating capacity factors for hydropower (dam and RoR) is detailed in Ref. [114]. The sustainable and economic hydropower potential is obtained from Ref. [30]. Geothermal energy potential is estimated according to the method described in Ref. [115]. The potential of the sustainable bioenergy is discussed in section 3.1.

3.5. Financial and technical assumptions

The modelling of the African power sector is done based on assumed financial and technical parameters from 2020 to 2050 in 5-year time steps of all the technologies employed. Financial assumptions are mainly about cost, including capital expenditure (CAPEX) and operational expenditure (OPEX) from 2020 onwards. Supplementary Material (Tables A1 and A3) depicts the financial assumptions, while technical assumptions regarding storage facilities self-discharge rate, losses in power lines, and efficiencies are found in the Supplementary Material (Tables A2 and A4). Except for residential PV prosumers, which has its weighted average cost of capital (WACC) set to 4% owing to lower returns on investment requirements, all scenarios have 7% as the weighted average cost of capital. Based on [116,117], end-use electricity prices for all sectors for the base year 2020 were calculated according to Ref. [117], and it is presented in the Supplementary Material (Table A5). The RE upper and lower limits of the energy resources were calculated based on Bogdanov and Breyer [114] and Farfan and Breyer [118] respectively and are shown in the Supplementary Material (Table A6).

3.6. Applied scenarios

To investigate the value and impact of biomass in the African power sector, this study developed and analysed two scenarios which are:

- Best Policy Scenario without bioenergy (BPS-1): This scenario aims for 100% RE by 2050 without bioenergy in the entire African power sector.
- Best Policy Scenario with bioenergy (BPS-2): This scenario aims for 100% RE by 2050 with bioenergy within realistic limits for use in the power sector.

The main idea is to investigate the two scenarios: a scenario without bioenergy and with bioenergy to analyse the impact of biomass on the power systems with large shares of VRE resources. BPS-2 will be used to highlight the significant influence of bioenergy in the African power sector.

4. Results

The sustainable bioenergy potential in country resolution, total installed electricity capacity, generation of electricity, power system balancing options (storage technologies, grid utilisation, flexible biomass generation), levelised cost of electricity (LCOE), regional capacity overview in a 100% RE system, and CO₂ emissions, are analysed in this section.

4.1. Sustainable bioenergy potential of Africa

This section presents the results of the estimation of the sustainable bioenergy potential of Africa. The bioenergy potential is highly influenced by crop production rather than forest cover. Crop residues account for 61% of the total sustainable bioenergy potential, followed by wood and wood residues, biogas, and MSW with shares of 22%, 13%, and 5%, respectively. The countries with high potential for sustainable biomass in Africa are Nigeria (256.5 TWh), Egypt (163.6 TWh), Ethiopia

Table 2

Estimated sustainable bioenergy potential of Africa in 2015 as calculated according to Mensah et al. [53].

Country	Unit	Crop residues	Wood and forest residues	Biogas	MSW	Total potential
Algeria	[TWh]	38.55	10.08	3.64	2.37	54.63
Angola	[TWh]	21.11	3.82	2.91	0.86	28.69
Benin	[TWh]	12.2	4.33	1.45	0.71	19.01
Botswana	[TWh]	0.39	0.63	1.03	0.13	2.17
Burkina Faso	[TWh]	24.02	10.43	6.19	1.37	42.01
Burundi	[TWh]	6.17	4.51	2.71	0.38	13.76
Cameroon	[TWh]	35.81	7.47	3.22	1.72	48.23
Cape Verde	[TWh]	0.16	0.12	0.05	0.09	0.42
Central African Republic	[TWh]	2.29	1.84	1.95	0.13	6.22
Chad	[TWh]	13.29	6.02	18.76	0.41	38.47
Comoros	[TWh]	0.37	0.29	0.10	0.10	0.87
Congo	[TWh]	1.63	1.28	2.39	0.17	5.47
Congo, DR	[TWh]	35.72	58.83	2.52	2.21	99.27
Cote D'Ivoire	[TWh]	27.57	7.42	1.39	1.56	37.94
Djibouti	[TWh]	0.08	0.36	0.22	0.03	0.69
Egypt	[TWh]	131.29	12.34	12.51	7.42	163.55
Equatorial Guinea	[TWh]	0.31	0.43	0.03	0.03	0.81
Eritrea	[TWh]	1.32	0.95	2.45	0.11	4.82
Ethiopia	[TWh]	85.32	4.12	39.02	0.26	128.72
Gabon	[TWh]	1.09	1.05	0.09	0.05	2.28
Gambia	[TWh]	1.08	0.65	0.33	0.06	2.13
Ghana	[TWh]	31.82	14.19	1.82	0.59	48.43
Guinea	[TWh]	1.48	1.97	0.37	0.05	3.88
Guinea Bissau	[TWh]	14.32	8.78	2.85	0.33	26.28
Kenya	[TWh]	31.31	19.52	12.83	0.30	63.96
Lesotho	[TWh]	0.40	1.59	0.66	0.06	2.71
Liberia	[TWh]	1.76	6.00	0.17	0.13	8.06
Libya	[TWh]	4.19	0.90	0.76	0.51	6.37
Madagascar	[TWh]	15.42	12.28	4.99	2.33	35.03
Malawi	[TWh]	26.68	5.09	1.36	0.48	33.62
Mali	[TWh]	33.42	4.59	7.12	3.99	49.11
Mauritania	[TWh]	0.98	1.77	2.63	0.12	5.49
Mauritius	[TWh]	2.07	0.24	0.17	0.09	2.56
Mayotte	[TWh]	0	0	0	0	0.00
Morocco	[TWh]	51.08	4.65	6.25	1.12	63.10
Mozambique	[TWh]	13.48	11.61	1.44	0.17	26.70
Namibia	[TWh]	0.87	1.03	1.33	0.07	3.29
Niger	[TWh]	32.83	8.65	9.18	1.81	52.47
Nigeria	[TWh]	182.12	53.17	17.74	3.47	256.51
Rwanda	[TWh]	11.79	3.51	1.06	0.33	16.68
Sao Tome and Principe	[TWh]	0.21	0.10	0.01	0.01	0.32
Senegal	[TWh]	12.60	4.34	2.79	0.42	20.15
Seychelles	[TWh]	0.01	0.00	0.01	0.02	0.05
Sierra Leon	[TWh]	5.80	4.66	0.63	0.19	11.28
Somalia	[TWh]	1.52	11.81	9.92	1.16	24.42
South Africa	[TWh]	58.15	12.04	7.31	40.60	118.10
Sudan	[TWh]	33.03	11.44	20.78	1.78	67.02
Swaziland	[TWh]	2.54	1.11	0.32	0.03	4.00
Tanzania	[TWh]	79.24	19.94	13.29	3.05	115.53
Togo	[TWh]	3.06	3.42	0.51	0.50	7.50
Tunisia	[TWh]	17.92	2.82	1.77	0.82	23.33
Uganda	[TWh]	30.25	30.33	7.91	0.50	68.99
Zambia	[TWh]	11.12	17.84	1.83	0.42	31.21
Zimbabwe	[TWh]	4.98	5.97	3.05	0.86	14.87
Total	[TWh]	1156.58	422.34	245.74	94.41	1911.18

(128.7 TWh), South Africa (118.1 TWh), and Tanzania (115.5 TWh). High levels of biogas potential are recorded in countries with a high population of cattle, such as Ethiopia (39.0 TWh), Sudan (20.8 TWh), Chad (18.8 TWh), and Nigeria (17.7 TWh). South Africa accounts for 40.6 TWh (47.0%) of energy recovery from MSW, followed by Egypt and Mali with potentials of 7.4 TWh (8.6%) and 4.0 TWh (4.6%),

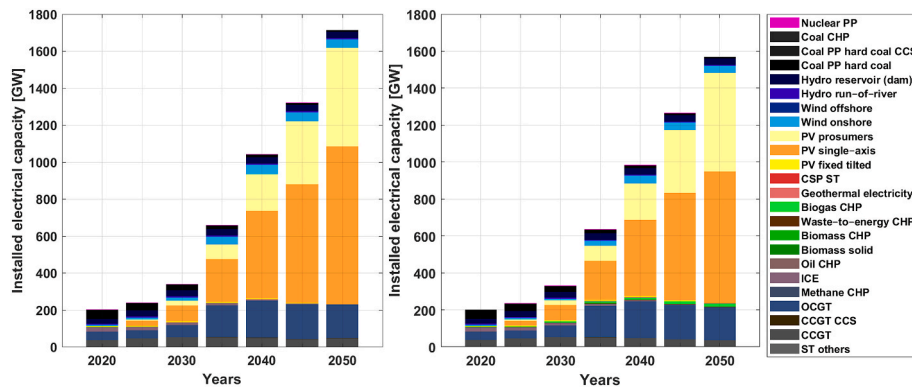


Fig. 5. Cumulative installed capacities in the BPS-1 (left) and BPS-2 (right) from 2020 to 2050. Abbreviations: ST - steam turbines, CCGT - combined cycle gas turbine, CCGT CCS - combined cycle gas turbine carbon capture storage, OCGT - open cycle gas turbine, CHP - combined heat and power, ICE - internal combustion engine, CSP ST - concentrated solar power steam turbine, PP - power plant.

respectively. A total of 1911.2 TWh of sustainable bioenergy potential is realised for the case of Africa. A share of 17.5% (335 TWh) of the total bioenergy (1911 TWh) harnessed in 2015 is applied to the African power sector for power supply using LUT-ESTM while the remaining share is reserved for the other sectors of the energy system. Table 2 below shows the sustainable bioenergy potential of Africa.

4.2. Electricity installed capacity

A least-cost transition to 100% RE was found for the BPS-1 and BPS-2, from a power sector based on high shares of fossil fuels and hydropower. The total installed capacities needed to satisfy the growing electricity demand in Africa from 2020 to 2050 is shown in Fig. 5. Supplementary Material (Tables A8–A10) also shows the capacity requirement during the transition period and shares of installed capacities by 2050. The contributions of various RE in the BPS-1 are, solar PV 1387 GW (80.9%), gas turbines 230 GW (13.4%), hydropower 41 GW (2.4%), wind power 46 GW (2.7%), coal 9 GW (0.5%), geothermal 1.44 GW (0.1%), CSP 0.38 GW (0%) and bioenergy 0 GW (0%), totalling 1714 GW. With bioenergy, in the BPS-2, these values are solar PV 1246 GW (79.4%), gas turbines 213 GW (13.6%), hydropower 41 GW (2.6%), wind power 37 GW (2.4%), bioenergy 22 GW (1.4%), coal 9 GW (0.5%), geothermal 1.40 GW (0.1%), and CSP 0.20 GW (0%) totalling 1570 GW. The total capacity instalment in the BPS-2 decreased by 8.40%. The capacity shares of dedicated renewable power capacities are 80% and 78.6% in BPS-1 and BPS-2 respectively by 2050, while the gas turbine capacities of 13.4% (BPS-1) and 13.6% (BPS-2) are operated only with renewable gas (biomethane, e-methane). The coal capacities are stranded in 2050 and non-operational. Capacity installations of geothermal and CSP could offer some flexibility. However, these technologies cannot compete economically with low-cost PV-battery hybrid

and biomass technologies. By 2050, 22 GW capacity addition of bioenergy power plants in the BPS-2 reduced 8 GW (19.6%) and 141 GW (10.2%) of wind and solar PV capacities, respectively, compared to BPS-1. The total installation of gas turbines also decreases by 17 GW (7.4%) by 2050, if bioenergy power plants are introduced in the BPS-2.

4.3. Electricity generation

Solar PV provides over 80% of electricity generation in both scenarios by 2050. By 2050, solar PV provides 2902 TWh (84.3%), followed by hydro reservoir 178 TWh (5.2%), renewable gas 170 TWh (4.9%), wind energy 148 TWh (4.3%), hydropower run-of-river 36 TWh (1.0%), geothermal energy 9 TWh (0.3%) and bioenergy 0 TWh (0%), of the total generation in the BPS-1. With bioenergy, these values are solar PV 2573 TWh (81.3%), hydropower reservoir 178 TWh (5.6%), bioenergy 130 TWh (4.1%), renewable gas 123 TWh (3.9%), wind energy 115 TWh (3.6%), hydropower run-of-river 36 TWh (1.1%) and geothermal energy 9 TWh (0.3%) in the BPS-2. By 2050, the total generation of electricity in the BPS-1 and the BPS-2 are 3443.3 TWh and 3164.1 TWh, respectively. The total generation in the BPS-2 decreased by 8.6% (299 TWh) in relation to BPS-1. Bioenergy impact is evident in the power generation as it supplies 130 TWh of electricity by 2050 from 22 GW of capacity instalment in the BPS-2. The higher electricity generation in the BPS-1 is due to a high share of inflexible solar PV and wind energy. The excess power goes to storage, e-methane, and curtailment. The electricity generation of both scenarios is shown in Fig. 6. More information on generation can be found in the Supplementary Material (Table A 11, Figure A4 – A6).

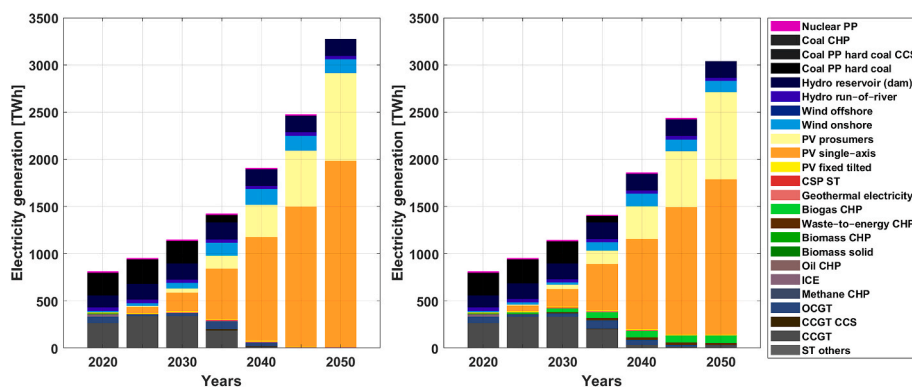


Fig. 6. Technology-wise electricity generation mix in the BPS-1 (left) and BPS-2 (right) from 2020 to 2050.

4.4. Role of power system balancing options

In Africa, inflexible generation from solar PV will be significant in the generation mix, making it necessary for the services of different balancing options to balance demand and supply of power in the system. Different balancing options are available to ensure day-to-day and seasonal balancing across the continent. These balancing options include resource complementarity, flexible generation, power grid infrastructure, curtailment, and storage.

4.4.1. Resource complementarity

Although Africa has a good potential of VRE, there is no direct investment competition expected between these resources as they will complement each other and not compete. Gerlach et al. [119], investigated the complementary versus competitive characteristics of solar and wind power resources by using reprocessed data provided by the German Aerospace Center (DLR). The results indicate that the FLH of the two major renewable power sources does not critically overlap over months and years in many regions of the world. For most regions in the world, the overlap, and the critical overlap FLH are about $15\% \pm 3\%$ and $2\% \pm 1\%$, respectively. The highest critical overlap in Africa is found in the Horn of Africa region [119]. Thus, there are negligible few occasions when both resource availability is excellent at the same time.

Often, when solar resource is excellent, wind resource is weak and vice versa. This complementary characteristic of solar and wind resources smoothens their variability, support each other in market diffusion, leads to a decrease in ramping requirements, decrease in capacity reserve, and increases the VRE penetration rate. Similarly, Solomon et al. [120,121] and Ludwig et al. [122] confirmed this complementary phenomenon between solar and wind resources. Additionally, other dispatchable RE resources such as hydropower, sustainable biomass, and geothermal energy further enhance the energy system resilience by providing flexible generation to complement solar and wind generation.

4.4.2. Flexible generation

Flexible generation from biomass provides a valuable balancing effect across Africa. In the BPS-2, when bioenergy is introduced into the generation mix, biomethane partly substitutes e-methane for gas turbine operations when generation from VRE resources is low, as depicted in Fig. 7. Flexible RE generation is mainly provided by bioenergy power plants of 130 TWh and hydropower reservoirs of 178 TWh for balancing local generation shortfalls across the African continent.

Flexible biomass generation in local demand areas affects grid utilisation, as shown in Fig. 8. The short-term and the long-term nature of transmission grid operation is evident, with greater use of the transmission grid in the BPS-1 compared to BPS-2. This is confirmed by the

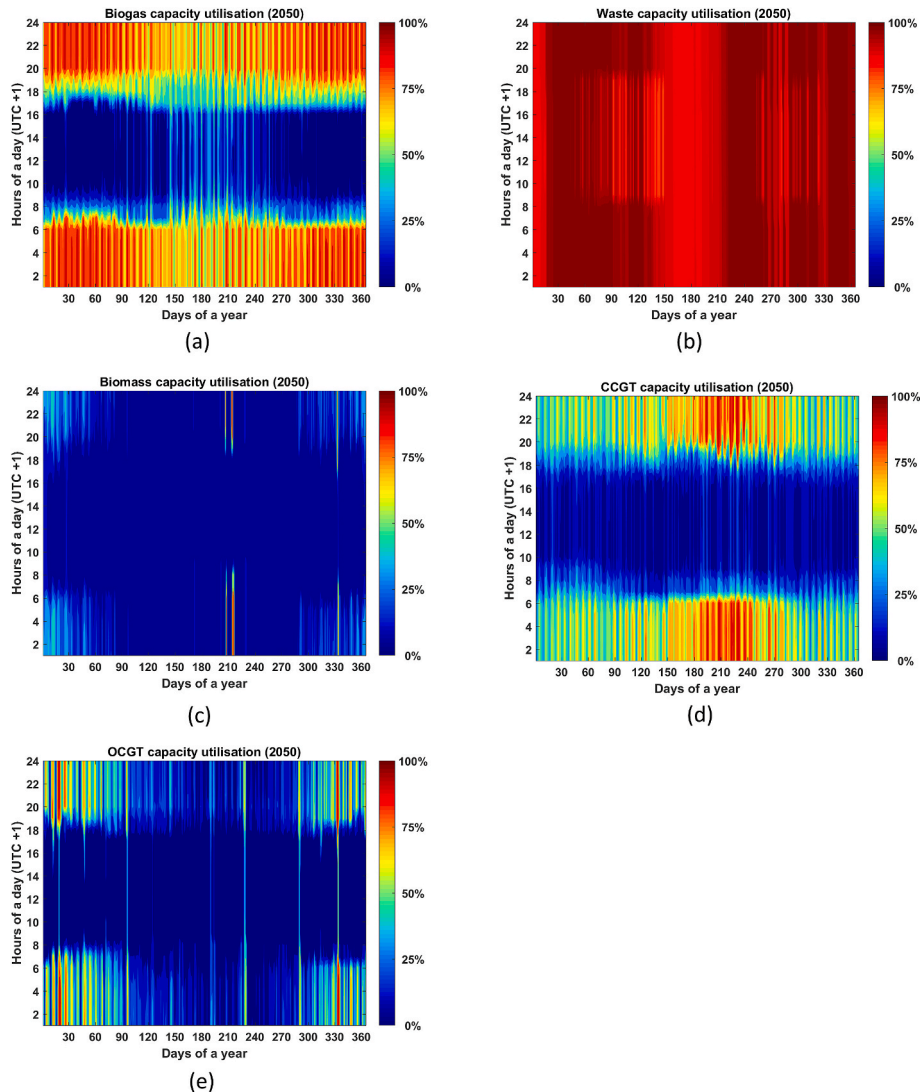


Fig. 7. Biogas utilisation (a), waste utilisation (b), solid biomass utilisation (c), CCGT gas turbine utilisation (d), and OCGT gas turbine utilisation (e) in the BPS-2 by 2050.

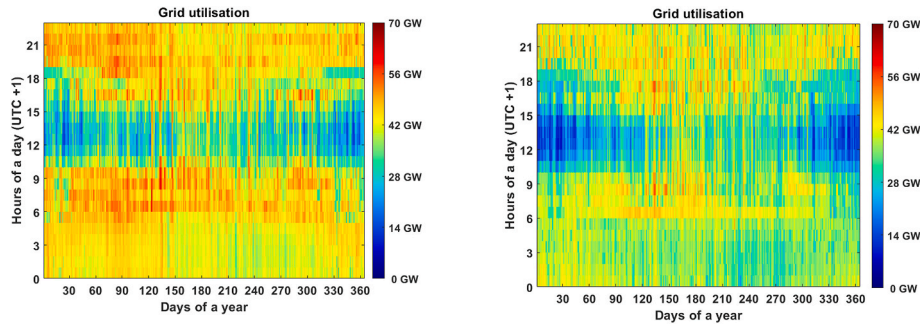


Fig. 8. Grid utilisation in the BPS-1 (left) and BPS-2 (right) by 2050.

Annual imported and exported electricity

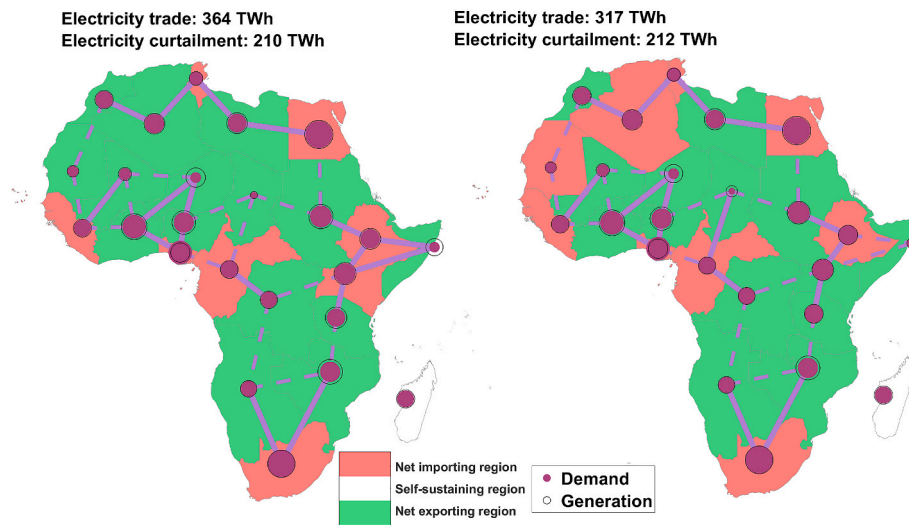


Fig. 9. Power exchange across Africa in the BPS-1 (left) and BPS-2 (right) by 2050.

reduction of the grid utilisation in the BPS-2 by 47 TWh (12.9% decrease). The difference of 47 TWh of traded electricity in Fig. 9 highlights the role of biomass in reducing grid utilisation due to its potential of satisfying local demand, thereby reducing the volume of power trade between the various regions.

4.4.3. Electricity import/export

The transmission and distribution (T&D) infrastructure are essential for any power sector development plan to achieve the universal access to electricity targets [94]. Grid interconnections enhance extra flexibility of the power sector between the regions. The electricity trade between various regions is illustrated in Fig. 9. A total electricity of 317 TWh is

exchanged between generating regions such as DZ, MA, LY, MLI, SE, SER, NER, NGA-N, MW, TCD, SOMDJ, COG, KENUG, SW, and WS and demand regions such as EG, ZAFLS, WW, TN, TZRB, NGA-S, ETH, and WW via the transmission grid to complement local generation by 2050 in the BPS-2. A solid line between regions shows power export while a dashed line depicts power import between connected regions. Comparatively, a total electricity of 364 TWh is exchanged between generating and demand regions in the BPS-1. It is empirically evident that less power exchange (a decrease of 12.9%) is experienced in the BPS-2 due to the biomass balancing effect in the local demand zones compared to the BPS-1.

Table 3

Storage units discharge in the BPS-1 and BPS-2 from 2020 to 2050.

Storage technologies	Unit	Scenarios	2020	2025	2030	2035	2040	2045	2050
Battery	TWh	BPS-1	0.0	1.5	22.2	153.0	490.0	743.9	1320.0
	TWh	BPS-2	0.0	1.1	17.1	127.7	426.2	680.6	1016.3
PHES	TWh	BPS-1	0.8	1.2	0.7	1.5	1.4	1.4	2.0
	TWh	BPS-2	0.8	1.2	0.8	1.5	1.4	1.4	1.9
TES	TWh	BPS-1	0.0	3.2	3.7	6.2	12.4	11.5	336.4
	TWh	BPS-2	0.0	3.3	3.2	5.8	11.9	13.0	10.4
DH TES	TWh	BPS-1	0.0	0.0	0.0	0.0	0.2	0.4	310.8
	TWh	BPS-2	0.0	0.1	0.0	0.2	0.5	6.2	8.7
A-CAES	TWh	BPS-1	0.0	0.0	0.0	0.0	0.0	0.0	27.5
	TWh	BPS-2	0.0	0.0	0.0	0.0	0.0	0.3	0.7
Gas (CH ₄)	TWh	BPS-1	0.0	0.0	0.0	0.0	0.2	1.8	68.8
	TWh	BPS-2	0.0	31.1	55.7	78.8	77.1	68.0	53.8

4.4.4. Curtailment

Another flexibility option is curtailment. It is the loss of generation during periods of low demand and high generation from VRE. Also unexpected mismatch in the system is managed by generation curtailment. Curtailment and storage capacity requirement are shown to substitute each other [123], and it is typically lower in cost to accept some curtailment, which typically increases as VRE penetration increases [123]. Curtailment information can be found in the Supplementary Material (Fig. A7). Other transition studies [124,125] have also shown that curtailment increases with increase in VRE penetration share. According to Li et al. [126], storage capacity has the tendency of reducing curtailment and vice versa. The results of this study show generation curtailment of 210 TWh (8.3% of generation) and 212 TWh (8.3% of generation) in the BPS-1 and BPS-2 respectively as depicted in Fig. 9.

4.4.5. Storage

Storage units are valuable for balancing of the power system since solar energy is not available at night and wind resource is less predictable in the long term. Table 3 depicts the storage discharge of the two scenarios. The total storage discharge declined by 47.1% from 2066 TWh to 1092 TWh in BPS-1 and BPS-2, respectively.

Battery systems dominate the storage output in both scenarios. The utility-scale battery takes the leading role from 2035 until 2050. The battery system electricity throughput of 1320 TWh and 1016 TWh in the BPS-1 and BPS-2, representing 63.9% and 93.1% of total storage discharge, respectively, by 2050, is higher than any other storage technology. This is because batteries are operated daily, resulting in about 454 and 358 full charge cycles per year in the BPS-1 and BPS-2, for an energy-to-power ratio of 6.37 h and 6.79 h, respectively.

Flexible generation from biomass and hydropower reservoir plants usually reduce the need for storage technologies. As observed in the BPS-2, the introduction of flexible bioenergy generation leads to a dramatic decrease in storage output. Battery system storage output decreased from 1320 TWh to 1016 TWh in BPS-1 and BPS-2. Whereas individual thermal energy storage (TES) decreased from 336 TWh to 10 TWh, district heating thermal energy storage (DH TES) storage decreased from 311 TWh to 9 TWh, A-CAES decreased from 28 TWh to 1 TWh, and e-methane output also decreased from 69 TWh to 54 TWh in the BPS-1 and BPS-2, respectively, by 2050. Extra storage output of 974 TWh is needed to compensate for the absence of flexible generation from biomass in the BPS-1. The total electricity output of the storage system is 1020 TWh (32% of generation) and 1381 TWh (40% of generation) in the BPS-2 and BPS-1, respectively.

It is permitted to install gas turbines from 2020 onwards since they can switch their fuel from fossil to renewable gas, i.e., either biomethane or synthetic electricity-based e-methane. The reason for opting for e-methane as a bridging fuel is to ease the transition process, as most thermal power infrastructures in Africa are gas turbines using natural

gas as fuel. Natural gas and e-methane share almost the same chemical characteristics and could be stored, transported, and combusted in the same infrastructure without major holistic technical changes to the system. It is, therefore, cost-effective to use e-methane instead of hydrogen for power generation. A share of 21% of the biogas in the BPS-2 is treated and upgraded to biomethane for gas turbine operation. Gas storage in the BPS-1 stores e-methane produced from renewable electricity in hours of supply surplus, especially solar PV. Gas storage is needed for operating the gas turbines in the monsoon seasons, especially when solar radiation is low; hence solar PV output is equally low, as shown in Fig. 10. The FLH of gas turbines decreased from 4075 h in 2020 to 705 h and 561 h by 2050 in the BPS-1 and BPS-2, respectively. Many African countries currently rely heavily on fossil natural gas for bulk power generation. However, by 2050, gas turbines are mainly used as seasonal balancing and peaker components for very limited hours of the year because of the high flexibility of gas turbines and the high cost of seasonal storage.

In the BPS-1 117 TWh are discharged from battery storage systems and converted into longer term storable e-methane, while in the BPS-2 this is reduced by 94.6% to 6.3 TWh. The battery-to-PtG phenomenon, as also reported by Gulagi et al. [95] and Oyewo et al. [50], is sensitive to the availability of seasonally dispatchable bioenergy. In the Supplementary Material (Figs. A8–A13), additional information on storage is presented.

4.5. Capacities in various regions in a 100% renewable energy system

The region with the highest total installed capacity is EG (Egypt) followed by ZAFLS (South Africa and Lesotho), WS (Benin, Cote D'Ivoire, Burkina Faso, Togo, Ghana), SE (Malawi, Mozambique, Zambia, Zimbabwe, Swaziland), and NGA-N (Nigeria North), with installed capacities of 248 GW, 246 GW, 161 GW, 93 GW, 96 GW, respectively in the BPS-2. These values are 247 GW, 242 GW, 164 GW, 105 GW, and 103 GW for the BPS-1. Fig. 11 shows the RE installed capacities in the various regions by 2050. In the regions of EG (Egypt), DZ (Morocco), and ZAFLS (South Africa and Lesotho), the excellent potential of solar, wind, and hydropower (in the case of Egypt) drives down capacity instalment in the BPS-1 because of resource complementarity. This case is seen in regions with very good solar, wind, and hydro resources potential. Otherwise, the reverse is the case in the other regions where the availability of biomass in the BPS-2 reduced the installed capacities.

Solar PV plays a dominant role in capacity installation in all regions. EG (Egypt), the leading region with solar PV installation, contributes 187 GW (15.0%) to the total solar PV installation of Africa (1246 GW), followed by ZAFLS (South Africa and Lesotho) and WS (Benin, Cote D'Ivoire, Burkina Faso, Togo, Ghana) with 162.87 GW (13.1%) and 146 GW (11.8%) respectively in the BPS-2. The regions around central Africa

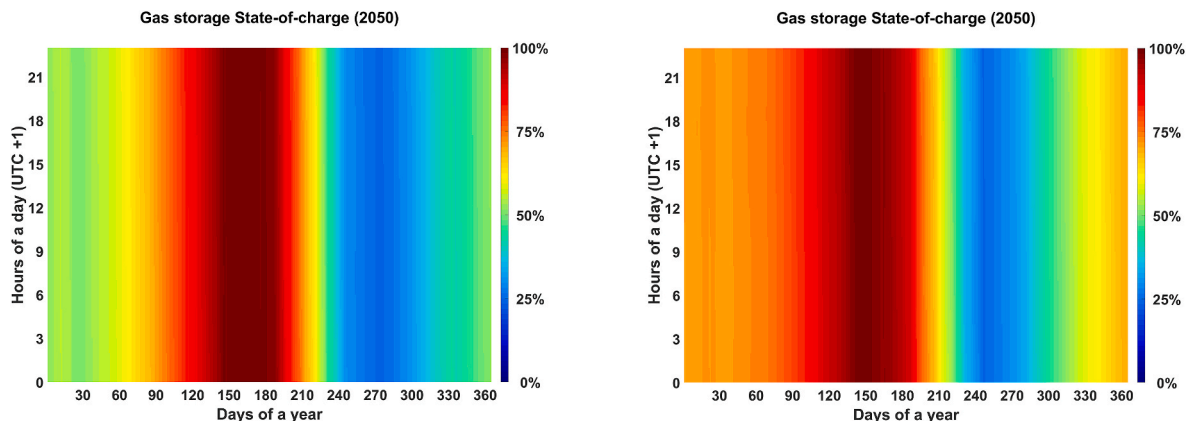


Fig. 10. Gas state of charge during an entire year in the BPS-1 (left) and BPS-2 (right) for 2050.

Regional electricity capacities

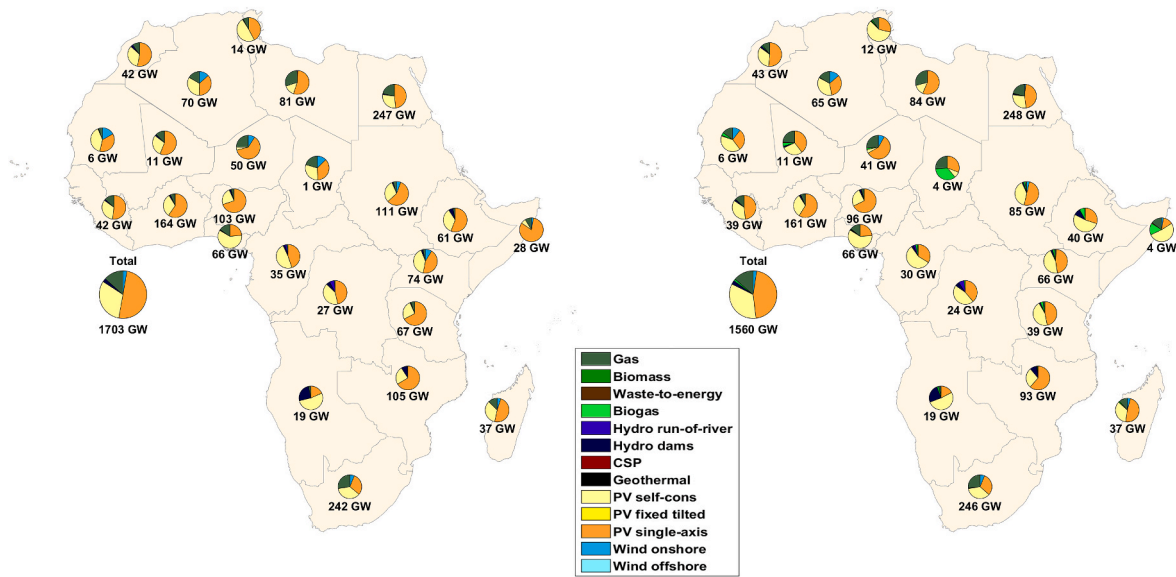


Fig. 11. Capacities installation in regional resolution for the BPS-1 (left) and BPS-2 (right) by 2050.

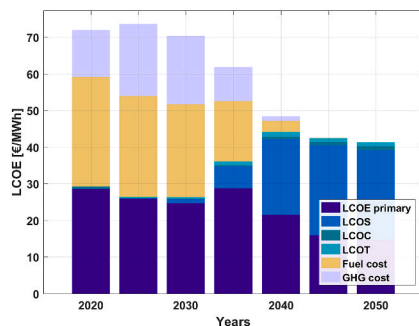
such as SW (Angola, Namibia, Botswana), COG (Congo, Democratic Republic), ETH (Ethiopia) and SE (Malawi, Mozambique, Zambia, Zimbabwe, Swaziland) and EG (Egypt) in North Africa contribute immensely to the total hydropower installed capacity of 41 GW by 2050 in both scenarios.

The regions with high share of biomass power installations are ETH (Ethiopia), ZAFLS (South Africa and Lesotho), EG (Egypt), KENUG (Kenya, Uganda), SER (Sudan, Eritrea), TZRB (Tanzania, Rwanda, Burundi), and TCD (Chad) with installed capacities of 2.85 GW (13.1%), 2.07 GW (9.5%), 1.75 GW (8.0%), 1.66 GW (7.6%), 1.62 GW (7.4%), 1.41 GW (6.5%), and 1.36 GW (6.3%) of the total biomass installation in Africa by 2050 in the BPS-2.

The dominant regions with high electricity generation by 2050 are EG (Egypt) followed by ZAFLS (South Africa and Lesotho), WS (Benin, Cote D'Ivoire, Burkina Faso, Togo, Ghana), SE (Malawi, Mozambique, Zambia, Zimbabwe, Swaziland), and NGA-N (Nigeria North) with a generation of 452 TWh, 408 TWh, 307 TWh, 234 TWh, and 205 TWh, respectively in the BPS-2. These values are 443 TWh, 387 TWh, 306 TWh, 259 TWh, and 215 TWh, respectively, in the BPS-1.

4.6. Levelised cost of electricity

Affordability of electricity affects its accessibility and utilisation [127]. The average cost of electricity is expressed as LCOE. The detailed



LCOE of both scenarios can be seen in Fig. 12. It includes the cost of fuel, curtailment, storage, CO₂ emission, transmission, and generation. By 2050, the LCOE in the BPS-1 and BPS-2 are 41.3 €/MWh and 37.1 €/MWh, respectively, decreasing from 71.9 €/MWh in 2020. In both scenarios, the cost of fuel and CO₂ emissions decrease from 2020 and finally disappear by 2050. Cost of storage and transmission declined substantially from BPS-1 to BPS-2 due to decrease in their respective installed capacities aided by biomass influence. Total annualised system cost declined from 110 b€ to 98.8 b€ in the BPS-1 and BPS-1, respectively. This dramatic decline is a result of reduced installed capacities of generation, transmission, and storage technologies. Two key observations are evident in the LCOE trajectory from 2020 to 2050. Firstly, in both scenarios, the LCOE increased sharply from 2020 to 2025, after which it declines till 2050. The slight initial increase in LCOE is due to initial investment requirements in RET. But the aftermath impact of low-cost RET causes the LCOE to decline from 2025 till 2050. Secondly, the difference of 4.2 €/MWh (a decrease of 10.1%) between BPS-1 and BPS-2 LCOE by 2050 highlights the positive impact of sustainable biomass on the overall system cost. More information on cost can be found in the Supplementary Material (Figs. A14–A18).

4.7. CO₂ emissions during the transition

Fig. 13 shows the CO₂ emission trajectory during the transition. The

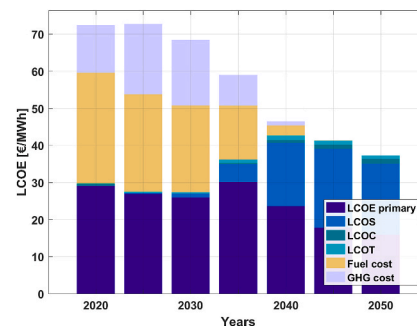


Fig. 12. Levelised cost of electricity in the BPS-1 (left), and BPS-2 (right) for 2050. Abbreviations: LCOE - primary levelised cost of electricity generation, LCOS - levelised cost of storage, LCOC - levelised cost of curtailment, LCOT - levelised cost of transmission.

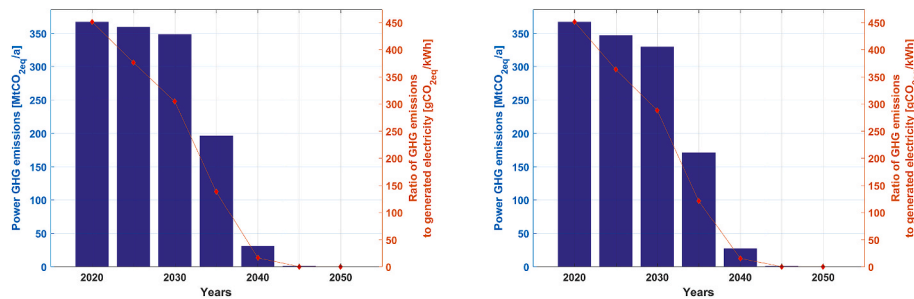


Fig. 13. The CO₂ emissions trajectory in the BPS-1 (left) and BPS-2 (right) during the transition period.

primary GHG emissions from fossil thermal power plants include CO₂, methane (CH₄), and nitrous oxide (N₂O) [97], with the dominance of CO₂. Both scenarios achieve zero emissions over the transition period. CO₂ emissions decrease from 367.2 MtCO_{2eq} in 2020 to 196.4 MtCO_{2eq} in BPS-1 and to 171.2 MtCO_{2eq} in BPS-2 by 2035, and by 2050, zero-emission is achieved in both scenarios, as shown in Fig. 13. The red curve shows the ratio of emitted CO₂ per kWh of electricity generated. The CO₂ emissions diminish from the system as fossil fuel technologies are utilised less and finally decommissioned during the transition period and replaced with RET capacities.

The BPS-2 starts recording lower levels of CO₂ emissions than the BPS-1 right from 2025 till both scenarios record zero by 2050. By 2035, deep decarbonization of 53.4% occurred in the BPS-2 compared to 46.5% in the BPS-1. The plausible reason for lower CO₂ emissions levels in the BPS-2 may be attributed to the presence of biomass (biogas) that is biogenic and therefore carbon neutral. The total cumulative pathway emissions are 5.78 GtCO₂ (BPS-1) and 5.47 GtCO₂ (BPS-2), which further highlights the positive impact of sustainable bioenergy use in faster CO₂ emission reduction.

5. Discussion

The transition towards a secure and more sustainable power system for Africa is made possible by RE generation, storage technologies, and transmission infrastructure between various regions of Africa. The results of the study indicate that RE resources can supply the total electricity demand of Africa by 2050, and that modern sustainable biomass will play an important role in overall system cost reduction and grid balancing. During the transition, LCOE values decreased from 71.9 €/MWh in 2020 to 41.3 €/MWh and 37.1 €/MWh in the BPS-1 and BPS-2 respectively by 2050. These LCOE values are comparable to other transition studies on some countries and parts of Africa, showing a range of 36.0–54.7 €/MWh for 2050 [42,56,60].

The decrease in LCOE over the transition period is mainly due to low-cost RE generation, especially solar and wind [26]. Solar PV is anticipated to have a decisive role in the future energy system of Africa [9,37,42]. Biomass power plants, together with hydropower reservoirs, are key flexible and dispatchable generators which are important resources and technologies to provide system balance and complement the variable nature of solar and wind generation in both short-term and long-term seasons. Furthermore, the total biomass potential of Africa is not exploited in this study. Only 17.5% (335 TWh) of the total potential is utilised. Sufficient supplies are reserved for the heating sector, biofuel production, transport, and biomass-to-chemicals in a full sector coupling scenario. The relative value of biomass in these sectors is expected to be higher than for power generation [87,128].

The CO₂ emissions reduced significantly as utilisation of fossil technologies decline and finally are replaced with low-cost RET. The results are in tangent with the United Nations Sustainable Development Goals (SDG), the Paris Agreement, and the AUC Agenda 2063 decision to provide clean energy for sustainable development in poorer countries [27,129].

The applied hierarchical modelling approach enabled the detailed modelling of Africa in 24 regions in full hourly resolution and power system interconnections. This approach may be expanded in future research from the two-level approach applied in this study to a four-level approach integrating more details. The first level could be on the level of major global regions as used in Ref. [38], using macro-regions as in this research on the second level, with meso-scale regions on the third level representing countries, groups of countries or parts of larger countries, and a fourth level on a national or sub-national level, as research on the case of Ethiopia [60], South Africa [56], Cameroon [59], Nigeria [54], Ghana [53], and West Africa [44]. Even a fifth level on a more local or municipal level may be applied. This would enable integrated modelling from a global to a local level.

A mix of e-methane and biomethane is employed in this study as a bridge fuel for power generation in the transition to a fully renewable power sector. It is imperative to state that the authors recognise the potential leakage of e-methane and biomethane from the system into the atmosphere due to fugitives and vented methane emissions [130]. This leakage is not accounted for in the study. The authors would therefore want to state unequivocally clear that the findings, conclusions, and recommendations regarding the use of e-methane and biomethane expressed in the paper may have to be further investigated in future research from a leakage perspective. In LUT-ESTM, a progressive effort is started to gradually phase out e-methane as a bridge fuel from the system. However, in the case of biomethane, it is rather of utmost benefit to the environment and society to convert waste and residues into biomethane in a more controlled energy system environment for power generation than disposing in landfills and dumpsites. Biological waste and residues which are disposed in landfills and dumpsites eventually get converted into methane via natural digestion. Biomethane leakages from energy recovering from waste processes are comparatively negligible to the tonnes of methane emissions released into the atmosphere from landfills and dumpsites due to natural digestion [131].

5.1. Significant role of bioenergy

Biomass can provide sustainable alternatives to fossil fuels [100] in addition to income generation and energy security for rural Africans. Regarding CO₂ emissions, using sustainable biomass as a substitute for fossil fuels in the African power system will be a beneficial policy initiative since biomass is regarded as carbon neutral. In particular if only sustainable biomass is used in the form of wastes, residues and by-products. Nevertheless, there must be careful planning, implementation, and social and environmental sustainability regulations to realize these benefits.

Over 90% of the forest wood harvest in Africa is used for fuel [132], and the United Nations Environmental Program (UNEP) forecasts that by 2050, 65% of the SSA population will still rely on wood fuel for cooking [133]. Currently, the SSA region relies significantly on biomass, and its growing population is expected to intensify harvesting of wood from forests [132], resulting in deforestation, GHG emissions, biodiversity loss and destruction of habitats [134]. It is important to regulate

forest harvest in Africa to curb deforestation by implementing forest protection policies [135]. The SSA region contributes 48.3% (2830 TWh) of the global total (5860 TWh) of traditional biomass for cooking [88]. Traditional biomass use for cooking is associated with low efficiency (10–20%) [87] and negative impact on human health due to high levels of particulate matter, CO₂, and other pollutants. According to the IEA, traditional biomass for cooking is responsible for the death of some 500 thousand Africans a year [86].

These problems could be curbed by transitioning from traditional biomass to sustainable modern biomass as defined in this study, which is beneficial for the African energy mix, while solar PV and battery storage is a real game changer. Residues from forests, crops and unavoidable waste and by-products from processes are used for electricity generation while applying a sustainability constrain on the forestry wood production. The results show that by 2050, 22 GW of biomass power plants are installed across the African continent generating some 130 TWh of electricity to satisfy local demand, thereby balancing the grid and enhancing a smooth integration of high shares of VRE.

One key problem in meeting the electricity needs of the SSA population is the widely disperse nature of rural communities, and grid connection would be rather costly due to transmission and distribution losses. Bertheau et al. [136] concluded that solar home systems (SHS) and mini-grids are the optimal solutions for isolated communities in SSA. Furthermore, Szabo et al. [137] also found that 36% of the un-electrified population in the SSA could be supplied with more affordable solar PV systems. It is in this context that decentralised electricity generation from local biomass and solar PV has been proposed as the means to fit this requirement eminently [138,139]. However, currently, only 2 GW of biomass capacity is installed in Africa. The key barriers to the introduction of modern biomass in the African electricity generation mix are firstly, the use of forest and crop residue in Africa for electricity generation is beset with the lack of awareness of matured biomass conversion technologies on the part of decision makers, users, and other stakeholders. Secondly, solid biofuels are usually not well treated to improve quality. They often contain high moisture contents (above 50%). Thirdly, biomass is very heterogeneous and complex; properties can vary with the species of plants, geographical location, growing condition, and storage. Fourthly, biomass has a relatively low energy content as compared to fossil fuels [140]. Fifthly, markets for bioenergy and other bioproducts are underdeveloped. Lastly, transportation to plant sites may be problematic due to the scattered nature of forest and crop residues, livestock manure, and MSW [67,140]. To successfully implement large-scale biomass power generation in Africa, these key hurdles must be overcome via strategic planning, implementation, and supervisions.

In the BPS-2 scenario, dispatchable power is mainly provided by hydropower, bioenergy plants, and gas turbines. A substantial decrease of installed capacity, total generation, storage output, grid utilisation, and LCOE by 8.5%, 8.2%, 46.2%, 13.7% and 10.1% respectively in the BPS-2 is achieved when bioenergy is added to the generation portfolio.

5.2. The role of hydropower

Hydropower is currently the primary source of renewable electricity, accounting for 16% and 22% of the total global and African electricity generation, respectively, in 2018 [34]. Hydropower accounts for over 39% and 50% of the total power installed capacity and supply in most SSA countries respectively [141]. Africa's hydropower potential is not fully exploited as 92% of the technically feasible potential remains unexploited [30]. Central, East, and Southern Africa holds the highest hydropower potential [142], with the Congo basin accounting for 40% of the continent's potential. However, several African countries relying on hydropower for power generation experiences frequent outages and power rationing due to prolonged droughts, and this may frequently occur in the future according to several studies [143–147].

Furthermore, large hydropower plant cost and schedule overruns

[148–150], environmental and societal impact [151], vulnerability to climate variability [145], makes it less competitive to low-cost solar and wind. Therefore, in this study, hydropower plants serve as a gap-filling technology, as also concluded by Oyewo et al. [43]. Similarly, Falchetta et al. [141] concluded that Africa would have to diversify its electricity generation portfolio to strengthen the power sector resilience. Hydropower can support a strong diversification of African power generation [152]. This study shows that a good mix of energy resources improves system flexibility, which is similar to the findings of Oyewo et al. [60] and Sterl et al. [152]. It is worth mentioning that the supply-side flexibility of the African power system is also linked to the flexibility of dammed hydropower plants. This study shows that hydropower supplies 214 TWh in both scenarios accounting for about 7% of the total power supply in a fully renewable power sector. Supply from hydropower reservoir (178 TWh), bioenergy (130 TWh), and storage technologies are significant for balancing high shares of VRE. Oyewo et al. [43] have shown that synthetic inertia can compensate for lack of classical inertia in a fully RE power system for SSA. Hourly generation in the worst and best weeks regarding renewable electricity supply can be found in the Supplementary Material (Figs. A21–A24).

5.3. Limitation of the study

Energy recovery from MSW involves considerable waste reception, handling and preparation steps which include crane sorting, crushing of bulky waste items, separation of waste fractions, and mixing of waste [153]. Due to its heterogenous characteristics, MSW are thoroughly mixed to blend the different fractions of the MSW to ensure consistent energy input to the combustion chamber of the waste to energy plant [153]. In this study, a mass combustion treatment technology is adopted as stated in section 3.1. Typically, mass combustion technology needs two waste preparation steps such as sorting of bulky items and mixing of different fractions for efficient waste combustion. The cost of these waste handling and preparation steps has not been factored in the overall system cost calculation of this study. Thus, this is recognised as a limitation of the study.

6. Conclusions

A sustainable bioenergy potential estimation method and the LUT Energy System Transition Model are employed to estimate and model the African sustainable bioenergy potential and the power sector transition, respectively. The impact and role of bioenergy on the power sector in an hourly resolved transition from 2020 to 2050 is investigated. As observed in the results, a defossilised power sector for Africa is possible and achievable by 2050, and it is the least cost solution for Africa. Such a system is vital in achieving the objectives of the United Nations Sustainable Development Goals, the Paris Agreement, and the AUC Agenda 2063.

The African power system is projected to transition to 100% renewable energy, driven by domestic availability of excellent and low-cost renewable energy resources and highly overall attractive economics of a 100% renewable power system. The most crucial power sector technologies will be solar photovoltaics providing more than 80% of all electricity generation and battery storage, enabling a 24/7 electricity supply based on solar energy. These two central core technologies are supported by dispatchable hydropower reservoir plants and bioenergy plants. Other renewables may play a significant role in specific regions, such as wind energy in Algeria, Niger, Egypt, Sudan and South Africa and geothermal energy in Tunisia, Nigeria, and Kenya, but they remain below 5% and 1%, respectively for the whole of Africa.

The total sustainable bioenergy resource from wastes, residues and by-products is estimated to be around 1910 TWh, thereof 17.5% are expected to be used as a dispatchable source of electricity, whereas the remaining potential is assumed to create an even higher value for contributing to cover heat demand, transportation demand and as a raw

material for biomass-to-chemicals conversion.

Notably, flexible generation from sustainable modern biomass offers system complementarity to variable renewable energy resources, thus enhancing the integration of large shares of variable renewable energy resources onto the power system. Flexible generation from biomass also provides an extra benefit to the system such as:

1. Traditional biomass use in Africa can be transformed into modern biomass to generate electricity for power system balancing.
2. Levelised cost of electricity is projected to decrease from 71.9 €/MWh in 2020 to 41.3 €/MWh and 37.1 €/MWh in the BPS-1 and BPS-2, respectively, by 2050.
3. Grid utilisation is reduced by 12.9% when bioenergy is added to the generation portfolio.
4. Total installed capacity and overall generation are reduced by 8.4% and 8.6%, respectively, in the BPS-2 as compared to the BPS-1.
5. Biomass technologies in the BPS-2 accelerate the CO₂ emissions reduction trajectory and lead to 5.4% less CO₂ emissions for the entire pathway.
6. The contribution of bioenergy in the power system decreased the required storage output by 47.1% in 2050.

Achieving a substantial contribution of sustainable bioenergy to the African power system will need good supportive policies, incentives, and domestic innovations to surmount any hurdles against the successful use of modern biomass in the region.

CRediT authorship contribution statement

Theophilus Nii Odai Mensah: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Ayo-bami Solomon Oyewo:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Dmitrii Bogdanov:** Writing – review & editing, Software, Methodology. **Arman Aghahosseini:** Data curation. **Christian Breyer:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors would want to declare that this research work received no funding.

Acknowledgments

The authors would like to thank the LUT Solar Economy group for specific and LUT University for general support. The authors would also like to thank Upeksha Caldera for proofreading the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2024.121198>.

References

- [1] V. Quaschnig, Renewable Energy and Climate Change, John Wiley and Sons Ltd, West Sussex UK, 2010, <https://doi.org/10.1002/9781119994381>.
- [2] [UN] - United Nations Framework Convention on Climate Change, Climate change: impacts, vulnerabilities and adaptation in developing countries, New York, <https://bit.ly/3jfOTDt>, 2007.
- [3] A. Azzuni, C. Breyer, Global energy security index and its application on national level, *Energies* 13 (10) (2020) 2502, <https://doi.org/10.3390/en13102502>.
- [4] [IPCC] – Intergovernmental Panel on Climate Change, Renewable Energy Sources and Climate Change Mitigation: Special Report of the Intergovernmental Panel on Climate Change, Cambridge: Cambridge University Press, United Kingdom, 2011, pp. 161–950, <https://doi.org/10.1017/CBO9781139151153>.
- [5] H. Ritchie, M. Roser, Emissions by sector, sect. By sect. Where do glob. Greenh. Gas emiss. Come from?, <https://bit.ly/3lcsEr8>, 2016. (Accessed 19 January 2022).
- [6] K. Hansen, C. Breyer, H. Lund, Status and perspectives on 100% renewable energy systems, *Energy* 175 (2019) 471–480, <https://doi.org/10.1016/j.energy.2019.03.092>.
- [7] C. Breyer, S. Khalili, D. Bogdanov, M. Ram, A.S. Oyewo, A. Aghahosseini, A. Gulagil, A.A. Solomon, D. Keiner, G. Lopez, P.A. Østergaard, H. Lund, B. V. Mathiesen, M.Z. Jacobson, M. Victoria, S. Teske, T. Pregger, V. Pthenakis, M. Rauegi, H. Holttinen, U. Bardi, A. Hoekstra, B.K. Sovacool, On the history and future of 100% renewable energy systems research, *IEEE Access* 10 (2022) 78176–78218, <https://doi.org/10.1109/ACCESS.2022.3193402>.
- [8] S. Teske, T. Pregger, S. Simon, T. Naegler, High renewable energy penetration scenarios and their implications for urban energy and transport systems, *Curr. Opin. Environ. Sustain.* 30 (2018) 89–102, <https://doi.org/10.1016/j.cosust.2018.04.007>.
- [9] M.Z. Jacobson, M.A. Delucchi, M.A. Cameron, S.J. Coughlin, C.A. Hay, I. P. Manogaran, Y. Shu, A.-K. von Krauland, Impacts of green new deal energy plans on grid stability, costs, jobs, health, and climate in 143 countries, *One Earth* 1 (2019) 449–463, <https://doi.org/10.1016/j.oneear.2019.12.003>.
- [10] [IPCC] – Intergovernmental Panel on Climate Change, Special report on renewable energy sources and climate change mitigation, Summary for policymakers and technical summary, Geneva, <https://shorturl.at/kffhm>, 2011.
- [11] N. Bird, C. Watson, L. Schalteck, K. Keil, Climate finance fundamentals: regional briefing: sub-Saharan Africa, Washington DC, <https://cdn.odi.org/media/documents/12085.pdf>, 2017.
- [12] [PRB] - Population Reference Bureau, World Population Data Sheet, Washington DC, 2017, https://assets.prb.org/pdf17/2017_World_Population.pdf.
- [13] D. Claude, Countries and continents of the world, <https://bit.ly/38ODp3N>, 2021. (Accessed 29 May 2021).
- [14] [IEA] - International Energy Agency, World energy outlook 2020, Paris, <https://www.iea.org/reports/world-energy-outlook-2020>, 2020.
- [15] [IEA]-International Energy Agency, SDG7: data and projections, access to ectr Paris, <https://www.iea.org/reports/sdg7-data-and-projections/access-to-electricity>, 2020. (Accessed 5 May 2021).
- [16] A. Castellano, A. Kendall, M. Nikomarov, T. Swemmer, McKinsey & Company, Brighter Africa: the Growth Potential of the Sub-Saharan Electricity Sector, New York, United States, 2015, <https://mck.co/3sNn73d>.
- [17] [UN] - United Nations, World economic situation prospects, New York, United States (2019). <https://bit.ly/3h6g1lv>.
- [18] [WB] - World Bank, GDP per capita (current US \$) - sub-Saharan Africa, Washington DC, <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=ZG>, 2019.
- [19] [UN] - United Nations, Department of Economic and Social Affairs, World population prospects 2019 highlights, New York, United States, https://population.un.org/wpp/Publications/Files/WPP2019_Highlights.pdf, 2019.
- [20] S.T. Chen, H.I. Kuo, C.C. Chen, The relationship between GDP and electricity consumption in 10 Asian countries, *Energy Pol.* 35 (2007) 2611–2621, <https://doi.org/10.1016/j.enpol.2006.10.001>.
- [21] [WB] - World Bank, Promoting regional power interconnection in central africa, Washington DC, <https://www.worldbank.org/en/news/press-release/2020/06/16/promoting-regional-power-interconnection-in-central-africa>, 2020.
- [22] [WB] - World Bank, ECOWAS-regional electricity access project, Washington DC, <https://bit.ly/3ha3CgA>, 2018.
- [23] P.L. Lucas, A.G. Dagnachew, A.F. Hof, Towards universal electricity access in Sub-Saharan Africa: a quantitative analysis of technology and investment requirements, PBL Netherlands Environmental Assessment Agency, The Hague, 2017, <https://bit.ly/3dh7J9c>.
- [24] [IEA] - International Energy Agency, Africa energy outlook, Paris,, <https://doi.org/10.1787/g2120ab250-en>, 2018.
- [25] [UN] - United Nations, Framework convention on climate change, the Paris agreement, New York, https://unfccc.int/sites/default/files/resource/paris_agreement_publication.pdf, 2016.
- [26] A.S. Oyewo, S. Sterl, S. Khalili, C. Breyer, Highly renewable energy systems in Africa: rationale, research, and recommendations, *Joule* 7 (2023) 1437–1470, <https://doi.org/10.1016/j.joule.2023.06.004>.
- [27] [AU]-Africa Union, Agenda 2063, The Africa We Want, 2015. Addis Ababa, https://au.int/sites/default/files/documents/33126-doc-01_background_note.pdf.
- [28] E. Vartiainen, G. Masson, C. Breyer, D. Moser, E. Román Medina, Impact of weighted average cost of capital, capital expenditure, and other parameters on future utility-scale PV levelised cost of electricity, *Prog. Photovoltaics Res. Appl.* 28 (2020) 439–453, <https://doi.org/10.1002/pip.3189>.
- [29] [IGA] - International Geothermal Association, Geothermal outlook in East africa: perspectives for geothermal development, Bochum, <https://bit.ly/3do7xpl>, 2018.
- [30] D.E.H.J. Gernaat, P.W. Bogaart, D.P. van Vuuren, H. Biemans, R. Niessink, High-resolution assessment of global technical and economic hydropower potential, *Nat. Energy* 2 (2017) 821–828, <https://www.nature.com/articles/s41560-017-0006-y>.
- [31] [BMZ], Federal ministry for economic cooperation and development, The Renewable Energy Transition in Africa, Frankfurt (2020). <https://shorturl.at/EZe1G>.
- [32] S. Khalili, C. Breyer, Review on 100% renewable energy system analyses – a bibliometric perspective, *IEEE Access* 10 (2022) 125792–125834, <https://doi.org/10.1109/Access.2022.3221155>.
- [33] Greenpeace, Energy [r]evolution-A sustainable world energy outlook 2015, Amsterdam, www.greenpeace.org/international/Global/%0Ainternational/publications/climate/2015/Energy-Revolution-2015-Full.pdf, 2015.
- [34] [IEA] – International Energy Agency, World energy outlook 2019, Paris, <https://bit.ly/3Azaafi>, 2019.

- [35] S. Teske, T. Pregger, S. Simon, T. Naegler, J. Pagenkopf, Ö. Deniz, B. Van Den Adel, K. Dooley, M. Meinshausen, It is still possible to achieve the Paris climate agreement: regional, sectoral, and land-use pathways. <https://doi.org/10.3390/en14082103>, 2021.
- [36] C. Breyer, D. Bogdanov, A. Gulagi, A. Aghahosseini, L. Barbosa, O. Koskinen, M. Barasa, U. Caldera, S. Afanasyeva, M. Child, J. Farfan, P. Vainik, On the role of solar photovoltaics in global energy transition scenarios, *Prog. Photovoltaics* 25 (2017) 727–745. <https://onlinelibrary-wiley-com.ezproxy.cc.lut.fi/doi/epdf/10.1002/ptp.2885>.
- [37] D. Bogdanov, J. Farfan, K. Sadovskaia, A. Aghahosseini, M. Child, A. Gulagi, A. S. Oyewo, L.S.N.S. Barbosa, C. Breyer, Radical transformation pathway towards sustainable electricity via evolutionary steps, *Nat. Commun.* 10 (2019) 1077, <https://doi.org/10.1038/s41467-019-08855-1>.
- [38] D. Bogdanov, M. Ram, A. Aghahosseini, A. Gulagi, A.S. Oyewo, M. Child, U. Caldera, K. Sadovskaia, J. Farfan, L.S.N.S. Barbosa, M. Fasihi, S. Khalili, T. Traber, C. Breyer, Low-cost renewable electricity as the key driver of the global energy transition towards sustainability, *Energy* 227 (2021) 120467, <https://doi.org/10.1016/j.energy.2021.120467>.
- [39] M. Ram, J.C. Osorio-Aravena, A. Aghahosseini, D. Bogdanov, C. Breyer, Job creation during a climate compliant global energy transition across the power, heat, transport, and desalination sectors by 2050, *Energy* 238 (2022) 121690, <https://doi.org/10.1016/j.energy.2021.121690>.
- [40] M. Ram, A. Aghahosseini, C. Breyer, Job creation during the global energy transition towards 100% renewable power system by 2050, *Technol. Forecast. Soc. Change* 151 (2020) 119682, <https://doi.org/10.1016/j.techfore.2019.06.008>.
- [41] A.S. Oyewo, B. Dmitrii, A. Aghahosseini, T.N.O. Mensah, C. Breyer, Contextualizing the scope, scale, and speed of energy pathways toward sustainable development in Africa, *iScience* 25 (2022) 104965, <https://doi.org/10.1016/j.isci.2022.104965>.
- [42] M. Barasa, D. Bogdanov, A.S. Oyewo, C. Breyer, A cost optimal resolution for Sub-Saharan Africa powered by 100% renewables in 2030, *Renew. Sustain. Energy Rev.* 92 (2018) 440–457, <https://doi.org/10.1016/j.rser.2018.04.110>.
- [43] A.S. Oyewo, J. Farfan Id, P. Peltoniemi, C. Breyer, Repercussion of large scale hydro dam deployment: the case of Congo grand inga hydro project, *Energies* 11 (4) (2018) 972, <https://doi.org/10.3390/en11040972>.
- [44] A.S. Oyewo, A. Aghahosseini, M. Ram, C. Breyer, Transition towards decarbonised power systems and its socio-economic impacts in West Africa, *Renew. Energy* 154 (2020) 1092–1112, <https://doi.org/10.1016/j.renene.2020.03.085>.
- [45] A. Aghahosseini, D. Bogdanov, C. Breyer, Towards sustainable development in the MENA region: analysing the feasibility of a 100% renewable electricity system in 2030, *Energy Strategy Rev.* 28 (2020) 100466, <https://doi.org/10.1016/j.esr.2020.100466>.
- [46] D. Timmons, A. Dhunny, K. Elaehe, B. Havumaki, M. Howells, A. Khodaruth, A. Lema-Driscoll, M. Lollchund, Y. Ramgolam, S. Rughooputh, D. Surroop, Cost minimization for fully renewable electricity systems: a Mauritius case study, *Energy Pol.* 133 (2019). <https://bit.ly/2TN8b8o>.
- [47] D. Timmons, K. Elaehe, M. Lin, Microeconomics of electrical energy storage in a fully renewable electricity system, *Sol. Energy* 206 (2020) 171–180, <https://doi.org/10.1016/j.solener.2020.05.057>.
- [48] S. Bouckaert, P. Wang, V. Mazauric, N. Maïzi, Expanding renewable energy by implementing dynamic support through storage technologies, *Energy Proc.* 61 (2014) 2000–2003, <https://doi.org/10.1016/j.egypro.2014.12.061>.
- [49] M. Drouineau, E. Assoumou, V. Mazauric, N. Maïzi, Increasing shares of intermittent sources in Reunion Island: impacts on the future reliability of power supply, *Renew. Sustain. Energy Rev.* 46 (2015) 120–128, <https://doi.org/10.1016/j.rser.2015.02.024>.
- [50] N. Maïzi, V. Mazauric, E. Assoumou, S. Bouckaert, V. Krakowski, X. Li, P. Wang, Maximizing intermittency in 100% renewable and reliable power systems: a holistic approach applied to Reunion Island in 2030, *Appl. Energy* 227 (2018) 332–341, <https://doi.org/10.1016/j.apenergy.2017.08.058>.
- [51] S. Seloese, S. Garabedian, O. Ricci, N. Maïzi, The renewable energy revolution of reunion island, *Renew. Sustain. Energy Rev.* 89 (2018) 99–105, <https://doi.org/10.1016/j.rser.2018.03.013>.
- [52] S. Seloese, O. Ricci, S. Garabedian, N. Maïzi, Exploring sustainable energy future in Reunion Island, *Util. Pol.* 55 (2018) 158–166, <https://doi.org/10.1016/j.jup.2018.10.006>.
- [53] T.N.O. Mensah, A.S. Oyewo, C. Breyer, The role of biomass in sub-Saharan Africa's fully renewable power sector – the case of Ghana, *Renew. Energy* 173 (2021) 297–317, <https://doi.org/10.1016/j.renene.2021.03.098>.
- [54] A.S. Oyewo, A. Aghahosseini, D. Bogdanov, C. Breyer, Pathways to a fully sustainable electricity supply for Nigeria in the mid-term future, *Energy Convers. Manag.* 178 (2018) 44–64, <https://doi.org/10.1016/j.enconman.2018.10.036>.
- [55] I.T. Tambari, M.O. Dioha, P. Failler, Renewable energy scenarios for sustainable electricity supply in Nigeria, *Energy Clim. Chang.* 1 (2020) 100017, <https://doi.org/10.1016/j.egycc.2020.100017>.
- [56] A.S. Oyewo, A. Aghahosseini, M. Ram, A. Lohrmann, C. Breyer, Pathway towards achieving 100% renewable electricity by 2050 for South Africa, *Sol. Energy* 191 (2019) 549–565, <https://doi.org/10.1016/j.solener.2019.09.039>.
- [57] P. Ferreira, A. Lopes, G. Dranka, J. Cunha, Planning for a 100% renewable energy system for the Santiago Island, Cape Verde, *Int. J. Sustain. Energy Plan. Manag.* 29 (2020) 25–40, <https://doi.org/10.5278/ijsepm.3603>.
- [58] E. Nordman, A. Barrenger, J. Crawford, J. McLaughlin, C. Wilcox, Options for achieving Cape Verde's 100% renewable electricity goal: a review, *Isl. Stud. Journal.* 14 (1) (2019) 41–58, <https://doi.org/10.24043/isj.73>.
- [59] N.B. Manjong, A.S. Oyewo, C. Breyer, Setting the pace for a sustainable energy transition in central africa: the case of Cameroon, *IEEE Access* 9 (2021) 145435–145458, <https://doi.org/10.1109/ACCESS.2021.3121000>.
- [60] A.S. Oyewo, A.A. Solomon, D. Bogdanov, A. Aghahosseini, T.N.O. Mensah, M. Ram, C. Breyer, Just transition towards defossilised energy systems for developing economies: a case study of Ethiopia, *Renew. Energy* 176 (2021) 346–365, <https://doi.org/10.1016/j.renene.2021.05.029>.
- [61] C. Breyer, M. Fasihi, A. Aghahosseini, Carbon dioxide direct air capture for effective climate change mitigation based on renewable electricity: a new type of energy system sector coupling, *Mitig. Adapt. Strategies Glob. Change* 25 (2020) 43–65, <https://doi.org/10.1007/s11027-019-9847-y>.
- [62] T. Schinko, S. Bohm, N. Komendantova, E.M. Jamea, M. Blohm, Morocco's sustainable energy transition and the role of financing costs: a participatory electricity system modeling approach, *Energy. Sustain. Soc.* 9 (2019), <https://doi.org/10.1186/s13705-018-0186-8>.
- [63] A. Purkus, E. Gaweł, N. Szarka, M. Lauer, V. Lenz, A. Ortwein, P. Tafarte, M. Eichhorn, D. Thrän, Contributions of flexible power generation from biomass to a secure and cost-effective electricity supply—a review of potentials, incentives and obstacles in Germany, *Energy Sustain Soc* 8 (2018) 18, <https://doi.org/10.1186/s13705-018-0157-0>.
- [64] [IPCC] - Intergovernmental Panel for Climate Change, Renewable Energy Sources and Climate Change Mitigation Special Report of the Intergovernmental Panel on Climate Change, Geneva, 2012. https://www.ipcc.ch/site/assets/uploads/2018/03/RRRen_Full_Report-1.pdf.
- [65] [IEA] - International Energy Agency, Electricity generation by source, Africa. <https://www.iea.org/data-and-statistics/data-browser/?country=WEOAFRICA&fuel=EnergySupply&indicator=ElecGenByFuel>, 2018.
- [66] [IEA] - International Energy Agency, Total Energy Supply (TES) by Source, Africa, 2018. <https://www.iea.org/data-and-statistics/data-browser/?country=WEOAFRICA&fuel=EnergySupply&indicator=TPESbySource>.
- [67] S. Dasappa, Potential of biomass energy for electricity generation in Sub-Saharan Africa, *Energy Sustain. Dev.* 15 (2011) 203–213, <https://doi.org/10.1016/j.esd.2011.07.006>.
- [68] M. Bildirici, F. Ozaksoy, Woody biomass energy consumption and economic growth in sub-Saharan Africa, *Procedia Econ. Financ.* 38 (2016) 287–293, [https://doi.org/10.1016/S2212-5671\(16\)30202-7](https://doi.org/10.1016/S2212-5671(16)30202-7).
- [69] E.A. Ayamga, F. Kemausuor, A. Addo, Technical analysis of crop residue biomass energy in an agricultural region of Ghana, *Resour. Conserv. Recycl.* 96 (2015) 51–60, <https://doi.org/10.1016/j.resconrec.2015.01.007>.
- [70] D. Roobroeck, R. Hood-Nowotny, D. Nakubulwa, J.B. Tumuhairwe, M.J. G. Mwanjalolo, I. Ndawula, B. Vanlauwe, Biophysical potential of crop residues for biochar carbon sequestration, and co-benefits, in Uganda, *Ecol. Appl.* 29 (8) (2019) e01984, <https://doi.org/10.1002/eap.1984>.
- [71] T.H. Debelo, A. Beyene, E. Tesfahun, A. Getaneh, A. Gize, Z. Mekonnen, Fecal contamination of soil and water in sub-Saharan Africa cities: the case of Addis Ababa, Ethiopia, *Ecohydrol. Hydrobiol.* 18 (2018) 225–230, <https://doi.org/10.1016/j.ecohyd.2017.10.003>.
- [72] [WB] - World Bank. What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050, Urban Development, Washington, DC, 2018. <https://sh.orturl.at/z25fz>.
- [73] T. Tomić, D.R. Schneider, The role of energy from waste in circular economy and closing the loop concept – energy analysis approach, *Renew. Sustain. Energy Rev.* 98 (2018) 268–287, <https://doi.org/10.1016/j.rser.2018.09.029>.
- [74] W.A. Qazi, M.F.M. Abushammala, Multi-criteria decision analysis of waste-to-energy technologies, in: *Waste-to-Energy*, Academic Press, 2020, pp. 265–316, <https://doi.org/10.1016/B978-0-12-816394-8.00010-0>.
- [75] P. Stehlík, Contribution to advances in waste-to-energy technologies, *J. Clean. Prod.* 17 (2009) 919–931, <https://doi.org/10.1016/j.jclepro.2009.02.011>.
- [76] M. Pavlas, M. Tou, L. Bébar, P. Stehlík, Waste to energy - an evaluation of the environmental impact, *Appl. Therm. Eng.* 30 (2010) 2326–2332, <https://doi.org/10.1016/j.applthermaleng.2009.10.019>.
- [77] Y. Shi, Y. Wang, Y. Yue, J. Zhao, T. Maraseni, G. Qian, Unbalanced status and multidimensional influences of municipal solid waste management in Africa, *Chemosphere* 281 (2021) 130884, <https://doi.org/10.1016/j.chemosphere.2021.130884>.
- [78] D. Cudjoe, P.M. Acquah, Environmental impact analysis of municipal solid waste incineration in African countries, *Chemosphere* 265 (2021) 129186, <https://doi.org/10.1016/j.chemosphere.2020.129186>.
- [79] T.O. Somorina, S. Adesolab, A. Kolawole, State-level assessment of the waste-to-energy potential (via incineration) of municipal solid wastes in Nigeria, *J. Clean. Prod.* 164 (2017) 804–815, <https://doi.org/10.1016/j.jclepro.2017.06.228>.
- [80] N. Scarlat, V. Motola, J.F. Dallemand, F. Monforti-Ferrario, L. Mofor, Evaluation of energy potential of municipal solid waste from african urban areas, *Renew. Sustain. Energy Rev.* 50 (2015) 1269–1286, <https://doi.org/10.1016/j.rser.2015.05.067>.
- [81] H. Haberla, K.-H. Erba, F. Krausmann, A. Bondeaub, C. Lauka, C.M. Llerb, C. Plutzara, J.K. Steinberger, Global bioenergy potentials from agricultural land in 2050: Sensitivity to climate change, diets and yields, *Biomass Bioenergy* 35 (2011) 4753–4769, <https://doi.org/10.1016/j.biombioe.2011.04.035>.
- [82] G. Fischer, L. Schratzenholzer, Global bioenergy potentials through 2050, *Biomass Bioenergy* 20 (2001) 151–159, [https://doi.org/10.1016/S0961-9534\(00\)00074-X](https://doi.org/10.1016/S0961-9534(00)00074-X).
- [83] H. Yamamoto, F. Fujino, K. Yamaji, Evaluation of bioenergy potential with a multi-regional global-land-use-and-energy model, *Biomass Bioenergy* 21 (2001) 185–203, [https://doi.org/10.1016/S0961-9534\(01\)00025-3](https://doi.org/10.1016/S0961-9534(01)00025-3).

- [84] I. Garba, R. Bellingham, Energy poverty: estimating the impact of solid cooking fuels on GDP per capita in developing countries - case of sub-Saharan Africa, *Energy* 221 (2021) 119770, <https://doi.org/10.1016/j.energy.2021.119770>.
- [85] B. Wicke, E. Smeets, H. Watson, A. Faaji, The current Bioenergy production potential of semi-arid and arid regions in sub-Saharan Africa, *Biomass Bioenergy* 35 (2015) 2773–2786, <https://doi.org/10.1016/j.biombioe.2011.03.010>.
- [86] [IEA]-International Energy Agency, SDG7: data and projections, Access to Clean Cook. <https://www.iea.org/reports/sdg7-data-and-projections>, 2020. (Accessed 15 June 2021).
- [87] [IRENA] - International Renewable Energy Agency, Biomass for heat and power, technology brief, Abu Dhabi, <https://bit.ly/2U27708>, 2015.
- [88] D. Keiner, L.S.N.S. Barbosa, D. Bogdanov, A. Aghahosseini, A. Gulagi, S. Oyewo, M. Child, S. Khalili, C. Breyer, Global-local heat demand development for the energy transition time frame up to 2050, *Energies* 14 (2021) 3814, <https://doi.org/10.3390/en14133814>.
- [89] F. Musonda, M. Millinger, D. Thrän, Optimal biomass allocation to the German bioeconomy based on conflicting economic and environmental objectives, *J. Clean. Prod.* 309 (2021) 127465, <https://doi.org/10.1016/j.jclepro.2021.127465>.
- [90] [FAOSTAT]-Food and Agriculture Organization Statistics, Forestry Production and Trade Africa, 2015. <https://bit.ly/2CeV17U>. (Accessed 2 May 2021).
- [91] [FAOSTAT]-Food and Agriculture Organization Statistics, Crop Production Africa, 2015. <https://bit.ly/2q6A3l6>. (Accessed 15 May 2020).
- [92] A. Koopmans, J. Koppejan, Agricultural and Forest Residue - Generation, Utilization and Availability, Kuala Lumpur, 1997. <https://bit.ly/2G2ciEq>.
- [93] [FAOSTAT]-Food and Agriculture Organization Statistics, Live Animal Africa, 2015. <https://bit.ly/2n184Sa>. (Accessed 6 June 2021).
- [94] [USDA]-United States Department of Agriculture, Animal Manure Management, Natural Resources Conservation Service, 1995. <https://bit.ly/3AyMjOl>. (Accessed 25 June 2021).
- [95] [WB]-World Bank, Africa Population, Total, 2020. Washington DC, <https://data.worldbank.org/indicator/SP.POP.TOTL>. (Accessed 16 April 2021).
- [96] C. Rose, A. Parker, B. Jefferson, E. Cartmell, The characterization of feces and urine: a review of the literature to inform advanced treatment technology, *Crit. Rev. Environ. Sci. Technol.* 45 (2015) 1827–1879, <https://doi.org/10.1080/10643389.2014.1000761>.
- [97] [IEA] - International Energy Agency, The Biogas Handbook. Science, Production and Application, Woodhead Publishing, Paris, 2013. <https://bit.ly/37tbHlR>.
- [98] [UN]-United Nations, Environmental Statistics, Department of Economic and Social Affairs. <https://unstats.un.org/unsd/envstats/qindicators.cshml>, 2021. (Accessed 28 January 2021).
- [99] [IEA] - International Energy Agency, Global wood pellet industry and trade study 2017, Paris, <https://bit.ly/3xVIHEz>, 2017.
- [100] D. Bogdanov, C. Breyer, North-East Asian Super Grid for 100% renewable energy supply: optimal mix of energy technologies for electricity, gas and heat supply options, *Energy Convers. Manag.* 112 (2016) 176–190, <https://doi.org/10.1016/j.enconman.2016.01.019>.
- [101] S. Comello, S. Reichelstein, The emergence of cost effective battery storage, *Nat. Commun.* 10 (2019) 2038, <https://doi.org/10.1038/s41467-019-09988-z>.
- [102] N. Hartmann, O. Vöhringer, C. Kruck, L. Eltrop, Simulation and analysis of different adiabatic Compressed Air Energy Storage plant configurations, *Appl. Energy* 93 (2012) 541–548, <https://doi.org/10.1016/j.apenergy.2011.12.007>.
- [103] C.C. Garcia, Testing a GIS-Based Methodology for Optimal Location of PumpedStorage Power Plants in Norway, Norwegian University of Science and Technology, 2013. <https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/242424>.
- [104] X. Zhang, C. Bauer, C.L. Mutel, K. Volkart, Life Cycle Assessment of Power-to-Gas: approaches, system variations and their environmental implications, *Appl. Energy* 190 (2017) 326–338, <https://doi.org/10.1016/j.apenergy.2016.12.098>.
- [105] C. Breyer, M. Fasihi, C. Bajamundi, F. Creutzig, Direct air capture of CO₂: a key technology for ambitious climate change mitigation, *Joule* 3 (2019) 2053–2065. <https://doi.org/10.1016/j.joule.2019.08.010>.
- [106] K. Sadoskaia, D. Bogdanov, S. Honkapuro, C. Breyer, Power transmission and distribution losses – a model based on available empirical data and future trends for all countries globally, *Int. J. Electr. Power Energy Syst.* 107 (2019) 98–109, <https://doi.org/10.1016/j.ijepes.2018.11.012>.
- [107] A. Toktarova, L. Gruber, M. Hlusiak, D. Bogdanov, C. Breyer, Long term load projection in high resolution for all countries globally, *Int. J. Electr. Power Energy Syst.* 111 (2019) 160–181, <https://doi.org/10.1016/j.ijepes.2019.03.055>.
- [108] D. Stetter, Enhancement of the REMix Energy System Model: Global Renewable Energy Potentials, Optimized Power Plant Siting and Scenario Validation, University of Stuttgart, 2012. <https://bit.ly/2viuxPR>.
- [109] P.W. Stackhouse, C.H. Whitlock (Eds.), Surface Meteorology and Solar Energy (SSE) Release 6.0, NASA SSE 6.0, Earth Science Enterprise Program, National Aeronautic and Space Administration (NASA), 2008. Langley, <http://eosweb.larc.nasa.gov/sse/>.
- [110] P.W. Stackhouse, C.H. Whitlock (Eds.), Surface Meteorology and Solar Energy (SSE) Release 6.0 Methodology, NASA SSE 6.0, Earth Science Enterprise Program, National Aeronautic and Space Administration (NASA), 2009. Langley, <http://eosweb.larc.nasa.gov/sse/>.
- [111] S. Afanasyeva, D. Bogdanov, C. Breyer, Relevance of PV with single-axis tracking for energy scenarios, *Sol. Energy* 173 (2018) 173–191, <https://doi.org/10.1016/j.solener.2018.07.029>.
- [112] C. Breyer, D. Bogdanov, A. Aghahosseini, A. Gulagi, M. Child, A.S. Oyewo, J. Farfan, K. Sadoskaia, P. Vainikka, Solar photovoltaics demand for the global energy transition in the power sector, *Prog. Photovoltaics Res. Appl.* 26 (2018) 505–523, <https://doi.org/10.1002/ppp.2950>.
- [113] K. Verzano, Climate change impacts on flood related hydrological processes: further development and application of a global scale hydrological model, *Reports on Earth System Science* 71 (2009). Max-Planck-Institut für Meteorologie, University of Kasse, <https://bit.ly/3b0lR5>.
- [114] D. Bogdanov, C. Breyer, North-East Asian Super Grid for 100% renewable energy supply: optimal mix of energy technologies for electricity, gas and heat supply options, *Energy Convers. Manag.* 112 (2016) 176–190, <https://doi.org/10.1016/j.enconman.2016.01.019>.
- [115] A. Aghahosseini, D. Bogdanov, C. Breyer, A techno-economic study of an entirely renewable energy-based power supply for North America for 2030 conditions, *Energies* 10 (8) (2017) 1171, <https://doi.org/10.3390/en10081171>.
- [116] C. Breyer, A. Gerlach, Global overview on grid-parity, *Prog. Photovoltaics Res. Appl.* 21 (2013) 121–136, <https://doi.org/10.1002/ppp.1254>.
- [117] A. Gerlach, C. Breyer, Impact of financing cost on global grid-parity dynamics till 2030, in: 29th Eur. Photovolt. Sol. Energy Conf., Amsterdam, 2014. <https://bit.ly/39FAVms>.
- [118] J. Farfan, C. Breyer, Structural changes of global power generation capacity towards sustainability and the risk of stranded investments supported by a sustainability indicator, *J. Clean. Prod.* 141 (2017) 370–384, <https://doi.org/10.1016/j.jclepro.2016.09.068>.
- [119] A.K. Gerlach, D. Stetter, J. Schmid, C. Breyer, PV and wind power - complementary technologies, in: 30th ISES Bienn. Sol. World Congr. 2011, SWC 2011, Kassel, 2011, pp. 1972–1978. [10.4229/26thEUPVSEC2011-6CV.1.32](https://doi.org/10.4229/26thEUPVSEC2011-6CV.1.32).
- [120] A.A. Solomon, M. Child, U. Caldera, C. Breyer, Exploiting wind-solar resource complementarity to reduce energy storage need, *AIMS Energy* 8 (2020) 749–770, <https://doi.org/10.3934/ENERGY.2020.5.749>.
- [121] A.A. Solomon, D.M. Kammen, D. Callaway, Investigating the impact of wind-solar complementarities on energy storage requirement and the corresponding supply reliability criteria, *Appl. Energy* 168 (2016) 130–145, <https://doi.org/10.1016/j.apenergy.2016.01.070>.
- [122] D. Ludwig, C. Breyer, A.A. Solomon, R. Seguin, Evaluation of an onsite integrated hybrid PV-Wind power plant, *AIMS Energy* 8 (2020) 988–1006, <https://doi.org/10.3934/energy.2020.5.988>.
- [123] A.A. Solomon, D. Bogdanov, C. Breyer, Curtailment-storage-penetration nexus in the energy transition, *Appl. Energy* 235 (2019) 1351–1368, <https://doi.org/10.1016/j.apenergy.2018.11.069>.
- [124] G. Lopez, A. Aghahosseini, D. Bogdanov, T.N.O. Mensah, N. Ghorbani, U. Caldera, A.P. Riverob, J. Kisselb, C. Breyer, Pathway to a fully sustainable energy system for Bolivia across power, heat, and transport sectors by 2050, *J. Clean. Prod.* 293 (2021) 126195, <https://doi.org/10.1016/j.jclepro.2021.126195>.
- [125] R. Satymov, D. Bogdanov, C. Breyer, The value of fast transitioning to a fully sustainable energy system: the case of Turkmenistan, *IEEE Access* 9 (2021) 13590–13610, <https://doi.org/10.1109/ACCESS.2021.3050817>.
- [126] C. Li, H. Shi, Y. Cao, J. Wang, Y. Kuang, Y. Tan, J. Wei, Comprehensive review of renewable energy curtailment and avoidance: a specific example in China, *Renew. Sustain. Energy Rev.* 41 (2015) 1067–1079, <https://doi.org/10.1016/j.rser.2014.09.009>.
- [127] H. Winkler, A.F. Simoes, E.L. Rovere, M. Alam, A. Rahman, S. Mwakasonda, Access and affordability of electricity in developing countries, *World Dev.* 39 (2010) 1037–1050. <https://bit.ly/3yDe0wk>.
- [128] J.B. Binder, R.T. Raines, Fermentable sugars by chemical hydrolysis of biomass, *Proc. Natl. Acad. Sci. U.S.A.* 107 (2010) 4516–4521, <https://doi.org/10.1073/pnas.0912073107>.
- [129] [UN] - United Nations, Transforming our world: the 2030 Agenda for sustainable development, United States, New York, 2015. <https://sdgs.un.org/partnerships>.
- [130] E.A. Grubert, A.R. Brandt, Three considerations for modeling natural gas system methane emissions in life cycle assessment, *J. Clean. Prod.* 222 (2019) 760–767, <https://doi.org/10.1016/j.jclepro.2019.03.096>.
- [131] T. Kormi, N.B.H. Ali, T. Abichou, R. Green, Estimation of landfill methane emissions using stochastic search methods, *Atmos. Pollut. Res.* 8 (2017) 597–605, <https://doi.org/10.1016/j.apr.2016.12.020>.
- [132] [WWF] - World Wildlife Fund. Chapter 2: Forests and Energy, WWF Living Forests Report, Gland, Switzerland, 2011. <https://shorturl.at/r15gE>.
- [133] [UNEP] - United Nations Environmental Programme, Review of woodfuel biomass production and utilization in africa: a desk study, Nairobi, <http://hdl.handle.net/20.500.11822/28515>, 2019.
- [134] S. Larcom, T. Van Gevelt, A. Zabala, Precolonial institutions and deforestation in Africa, *Land Use Pol.* 51 (2016) 150–161, <https://doi.org/10.1016/j.landusepol.2015.10.030>.
- [135] [FAO], Food and Agriculture Organization of the United Nations, southern africa's forests and people, investing in a sustainable future, Paris, <http://www.fao.org/3/i4894e/i4894e.pdf>, 2015.
- [136] P. Bertheau, A.S. Oyewo, C. Cader, C. Breyer, P. Blechinger, Visualizing national electrification scenarios for sub-saharan African countries, *Energies* 10 (2017) 1899, <https://doi.org/10.3390/en10111899>.
- [137] S. Szabó, I.P. Pascua, D. Puig, M. Moner-Girona, M. Negre, T. Huld, Y. Mulugetta, I. Kougiass, L. Szabó, D. Kammen, Mapping of affordability levels for photovoltaic-based electricity generation in the solar belt of sub-Saharan Africa, East Asia and South Asia, *Sci. Rep.* 11 (2021) 3226, <https://doi.org/10.1038/s41598-021-82638-x>.
- [138] N. Nelson, J. Darkwa, J. Calautit, M. Worall, R. Mokaya, E. Adjei, F. Kemausuor, J. Ahiekpor, Potential of bioenergy in rural Ghana, *Sustainability* 13 (2021) 381, <https://doi.org/10.3390/su13010381>.

- [139] J.T. Lee, D.S. Callaway, The cost of reliability in decentralized solar power systems in sub-Saharan Africa, *Nat. Energy* 3 (2018) 960–968, <https://doi.org/10.1038/s41560-018-0240-y>.
- [140] [BEL] - Biorenewables Education Laboratory, Biomass Properties and Handling, 2011. https://www.engineering.iastate.edu/brl/files/2011/10/brl_biomass_ppop_instructor.pdf.
- [141] G. Falchetta, D.E.H.J. Gernaat, J. Hunt, S. Sterl, Hydropower dependency and climate change in sub-Saharan Africa: a nexus framework and evidence-based review, *J. Clean. Prod.* 231 (2019) 1399–1417, <https://doi.org/10.1016/j.jclepro.2019.05.263>.
- [142] M. Hafner, M. Tagliapietra, L. de Strasser, Prospects for renewable energy in africa, in: *Energy Africa Challenges Oppor.*, SpringerBriefs in Energy, Cham, Springer, 2018, pp. 47–75, https://doi.org/10.1007/978-3-319-92219-5_3.
- [143] V. Sridharan, O. Broad, A. Shivakumar, M. Howells, B. Boehlert, D.G. Groves, H.-H. Rogner, C. Taliotis, J.E. Neumann, K.M. Strzepek, R. Lempert, B. Joyce, A. Huber-Lee, R. Cervigni, Resilience of the Eastern African electricity sector to climate driven changes in hydropower generation, *Nat. Commun.* 10 (2019) 302, <https://doi.org/10.1038/s41467-018-08275-7>.
- [144] P. Stanzela, H. Klinga, H. Bauer, Climate change impact on West African rivers under an ensemble of CORDEX climate projections, *Clim. Serv.* 11 (2018) 36–48, <https://bit.ly/3pSUSPD>.
- [145] D. Conway, C. Dalin, W.A. Landman, T.J. Osborn, Hydropower plans in eastern and southern Africa increase risk of concurrent climate-related electricity supply disruption, *Angew. Chem. Int. Ed.* 6 (11) (1967) 946–953, <https://doi.org/10.1038/s41560-017-0037-4>, 951–952. 2.
- [146] S. Turner, Y.N. Jia, S. Galelli, Examining global electricity supply vulnerability to climate change using a high-fidelity hydropower dam model, *Sci. Total Environ.* 590–591 (2017) 663–675, <https://bit.ly/3zpaJd8>.
- [147] N.V. Emodi, T. Chaiechi, A.B.M. Rabiul, A. Beg, The impact of climate variability and change on the energy system: a systematic scoping review, *Sci. Total Environ.* 676 (2019) 545–563, <https://doi.org/10.1016/j.scitotenv.2019.04.294>.
- [148] A. Ansar, B. Flyvbjerg, A. Budzier, D. Lunn, Should We Build More Large Dams? the Actual Costs of Hydropower Megaproject Development, vol. 69, 2014, pp. 43–56, <https://doi.org/10.1016/j.enpol.2013.10.069>.
- [149] O. Awojobi, G.P. Jenkins, Managing the cost overrun risks of hydroelectric dams: an application of reference class forecasting techniques, *Renew. Sustain. Energy Rev.* 63 (2016) 19–32, <https://doi.org/10.1016/j.rser.2016.05.006>.
- [150] R.W. Bacon, J.E. Besant-Jones, Estimating construction costs and schedules Experience with power generation projects in developing countries, *Energy Pol.* 26 (1998) 317–333, [https://doi.org/10.1016/S0301-4215\(97\)00164-X](https://doi.org/10.1016/S0301-4215(97)00164-X).
- [151] D.M. Rosenberg, R.A. Bodaly, P.J. Usher, Environmental and social impacts of large scale hydro- electric development: who is listening? *Global Environ. Change* 5 (1995) 127–148, <https://www.sciencedirect.com/science/article/pii/095937809500018J>.
- [152] S. Sterl, D. Fadly, S. Liersch, H. Koch, W. Thiery, Linking solar and wind power in eastern Africa with operation of the Grand Ethiopian Renaissance Dam, *Nat. Energy* 6 (2021) 407–418, <https://doi.org/10.1038/s41560-021-00799-5>.
- [153] A. Tabasová, J. Kropáč, V. Kermes, A. Nemet, P. Stehlík, Waste-to-energy technologies: impact on environment, *Energy* 44 (2012) 146–155, <https://doi.org/10.1016/j.energy.2012.01.014>.